

Pulmonary Pathology

By Dr. Tewodros D. (MD, Pathologist)

- Diseases of the upper airways
 - The upper airway includes the nose, pharynx ,larynx & related structures.
- Nose
 - Inflammations - most common , viral
 - Destructive inflammatory lesions - less common
 - Tumors - primary nasal cavity and sinuses

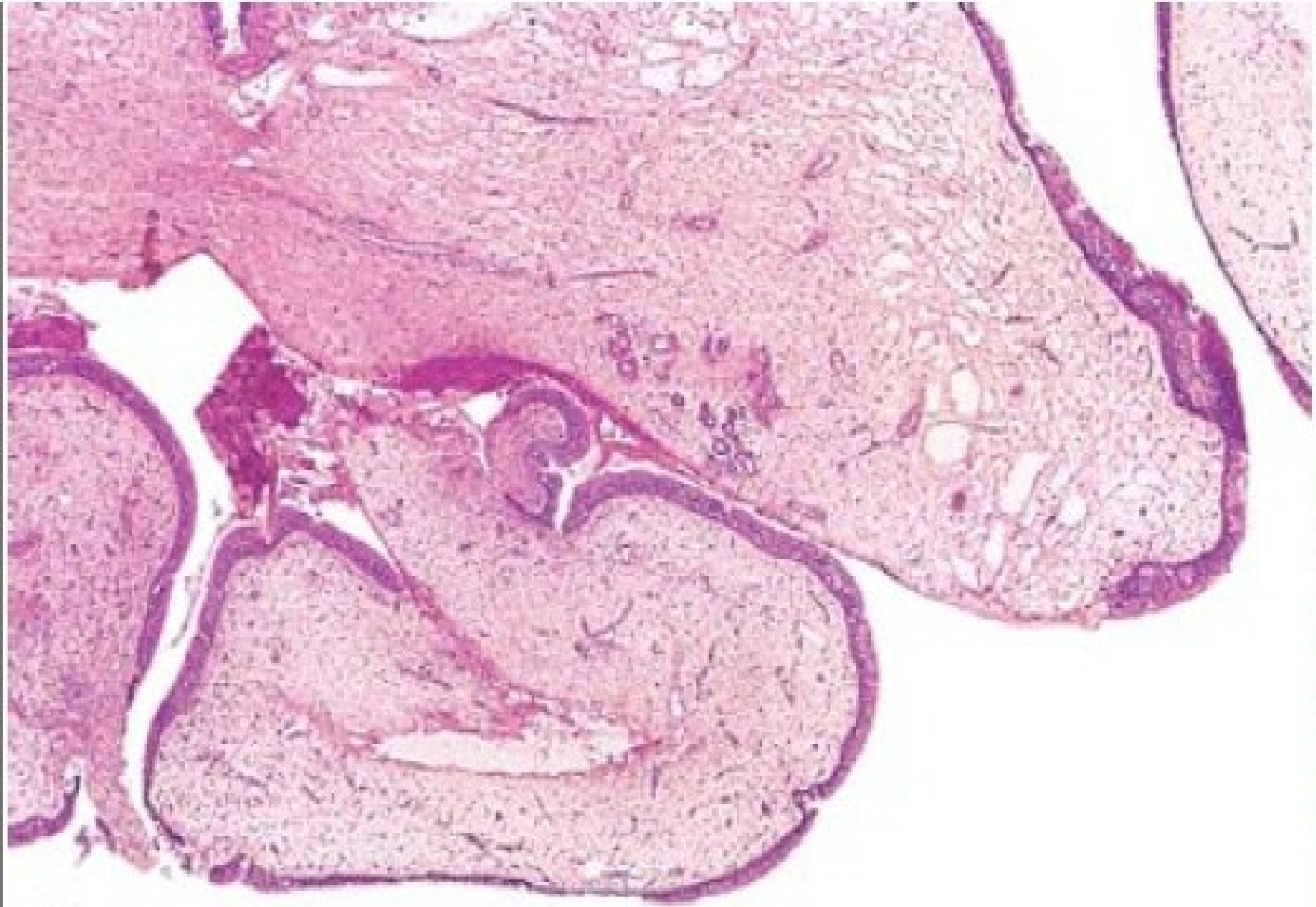
Inflammations

1. Infectious Rhinitis

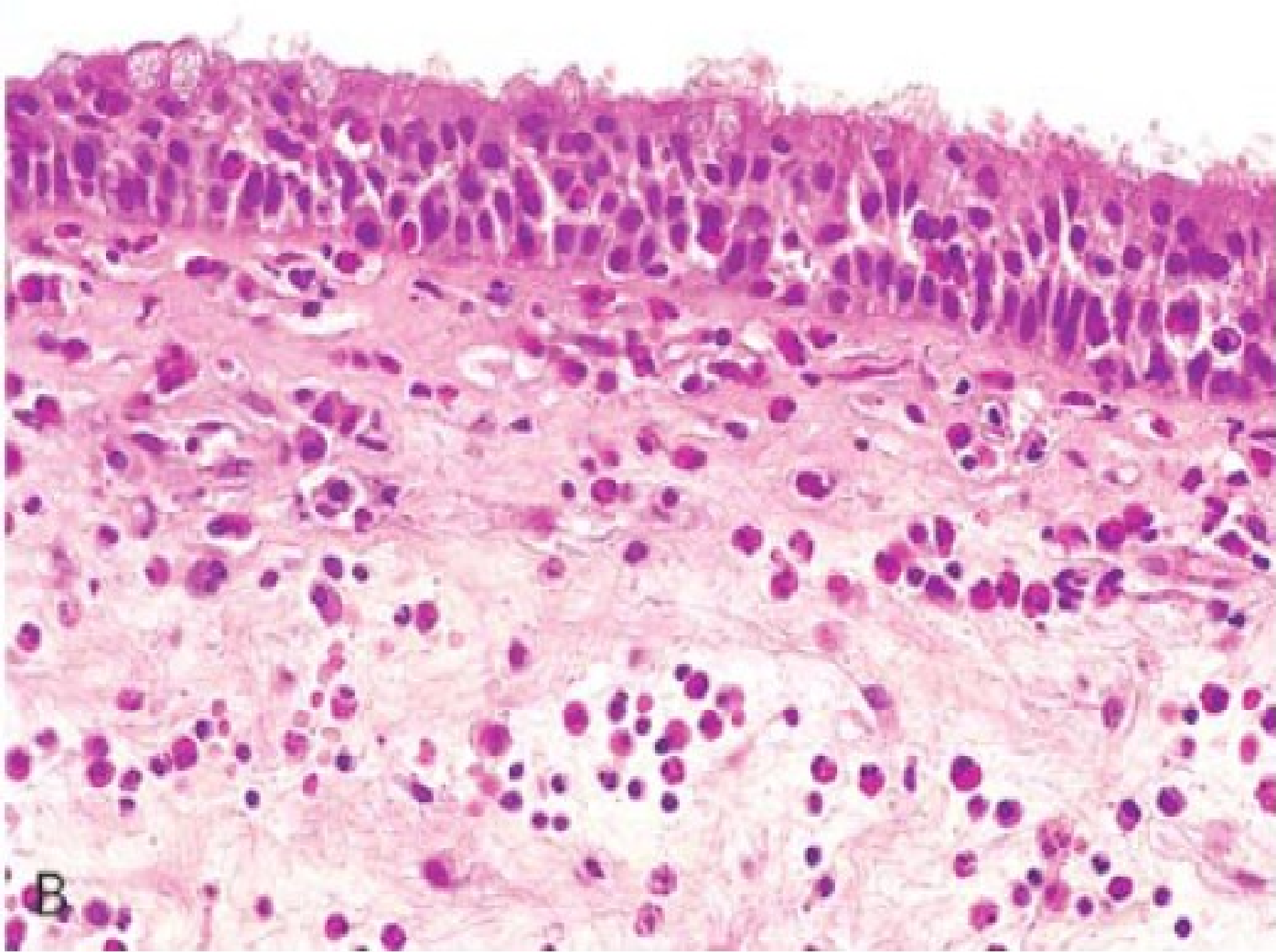
- Commonly called common cold
- Common offenders are adeno , rhino & echo viruses
- Evoke profuse catarrhal discharge
- Nasal mucosa - Thickened , edematous , red
- Narrowed cavities , enlarged turbinates
- Pharyngotonsillitis - concomitant
- Secondary bacterial infection - mucopurulent / suppurative exudates

2. Allergic Rhinitis (hay fever)

- It is an IgE mediated immune reaction
- Plant pollen ,fungi, animal allergens, dust mite. Early and late phase response.
- Mucosal edema, redness, mucus secretion , leukocytic infiltration (eosinophils)
- Nasal polyps
 - Recurrent attacks of rhinitis - focal protrusion of mucosa
 - Small , but 3 - 4 cm, Edematous mucosa
 - Stroma - loose , hyperplastic / cystic glands ,mixed inflammatory infiltrates
 - Ulcerate / infected , obstruct airway , impair sinus drainage



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3. Chronic rhinitis

- . Repeated attacks of acute rhinitis
- . Microbial / allergic
- . Superimposed bacterial infection
- . Impaired drainage / deviated septum

4. Sinusitis (acute)

- o Commonly preceded by rhinitis
- o Impairment of drainage of the sinus by inflammatory edema of the mucosa
- o Complete
 - empyema of sinus
 - mucocele
- o Spread to orbit , osteomyelitis ,
penetrate to cranial vault

- **Chronic sinusitis**
 - follow acute sinusitis
 - interference with drainage
 - mixed microbial flora
- * **sever forms caused by fungi (mucormycosis) esp. in DM**
- * **maxillary sinusitis : extension of peri apical tooth infection , mixed bacterial from oral mucosa.**

Necrotizing lesions of the nose and upper airways

- Necrotizing ulcerating lesions
 - Spreading fungal infections (mucormycosis) in the diabetic
- Wegner granulomatosis
- Lethal midline granuloma - tumor of NK cells (angio-centric NHL)



Inflammations

- Pharyngitis and tonsillitis
 - Frequent concomitants of viral infections
 - Bacterial infections may be superimposed / primary infection
 - Beta hemolytic strept -most common, *S. aureus*
 - Severe infection : infants/children , neutropenia , immunodeficiency , uncontrolled diabetes , disruption of oral flora by antibiotics
 - Major importance of streptococcal 'sore throat' lies in late sequel of rheumatic fever and glomerulo nephritis

Tumors of the nose, sinuses and nasopharynx

- Infrequent , include
 - Nasopharyngeal angiofibroma
 - Sinonasal papillomas
 - Isolated plasmocytomas
 - Olfactory neuroblastomas
 - Nasopharyngeal carcinoma

Nasopharyngeal carcinoma

- Keratinizing squamous cell carcinoma
- Non keratinizing SCC
- Undifferentiated carcinoma
- Three factors affect its origin
 - Heredity
 - Age -bimodal distribution, 15 -25 then 60-69
 - Infection with EBV
- Common in Africa & southeast Asia
- Mass - naso pharynx , tonsils, posterior tongue, upper airways

Spread and metastasis

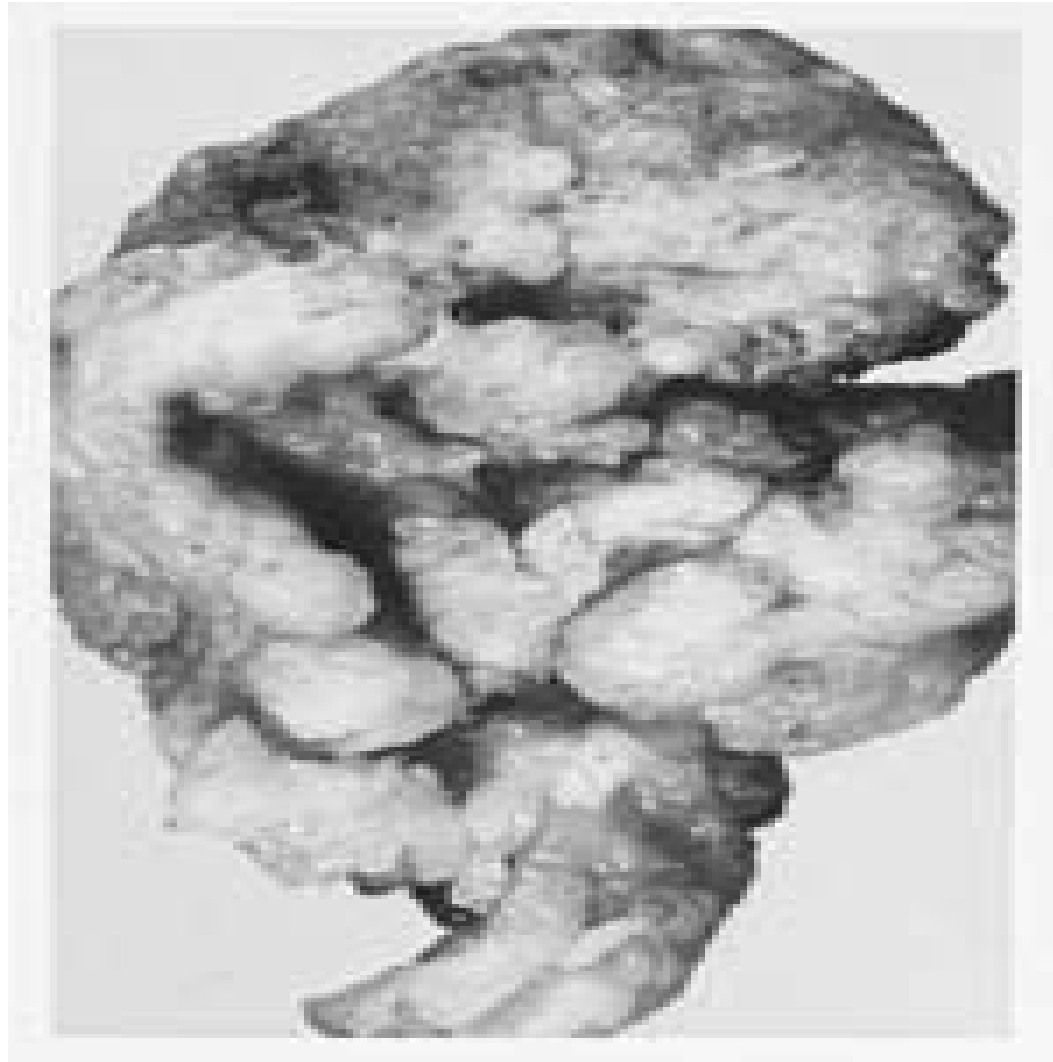
- Grow silently , unilateral cervical LAP commonest presentation
- Unrespectable , metastasized to local LNs and distant sites
- Radiotherapy - standard Tx , 50 - 70 % 3 year survival rate
- Undifferentiated ca - most sensitive , keratinizing the least sensitive

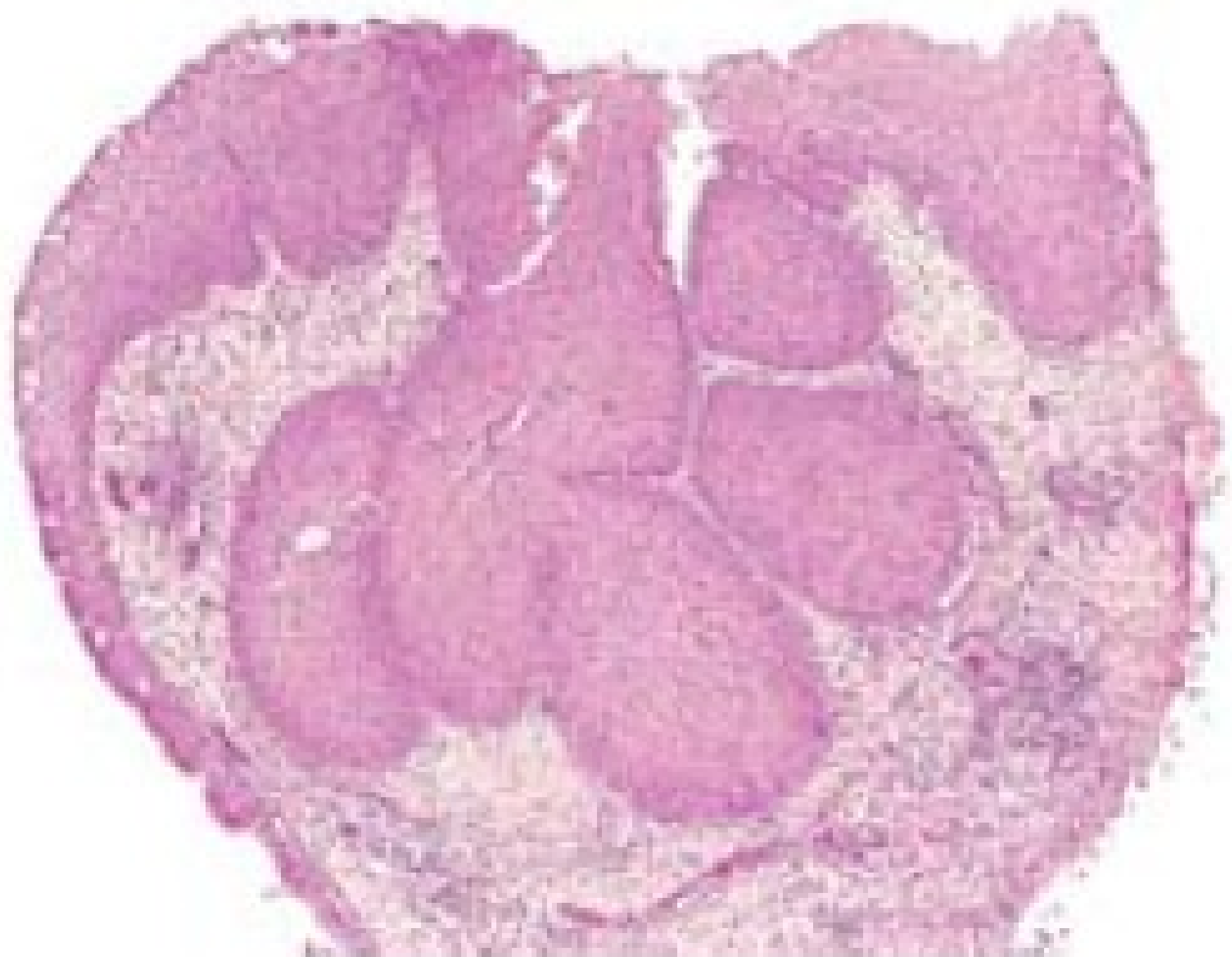
Nasopharyngeal angiofibroma

- Almost exclusively in males 10-25 yrs
- Posterio lateral wall of roof of nose , benign
- Polypoid mass , bleeds severely on manipulation and biopsy
- Obstruct nares, grow posteriorly , in to orbit / cranial vault
- Mic: mixture of fibrous and vascular tissue

Sinonasal papillomas

- Benign, sinonasal mucosa, squamous/columnar epith.
- Adult men , HPV types 6 & 11
- Stiffness, epistaxis ,obstruction
- Septal , inverted , cylindrical
- Inverted : locally aggressive , papilomatous proliferation of epithelium extends in to the mucosa ; High rate of recurrence , invade the orbit / cranial vault
- Septal /cylindrical : exophytic growth







Isolated plasmacytoma

- Extra medullary plasmacytoma
- Lymphoid tissue
- Polypoid growth , intact mucosa
- Soft bleeding mass
- 1cm - several cm
- Mic : Immature + mature plasma cells
- Rarely progress to multiple myeloma

Olfactory neuroblastoma

- Uncommon, highly malignant
- Composed of small round cells
- Arise from neuroendocrine cells in the olfactory mucosa
- 3 -79 yrs , mean 50 yrs
- Reddish polypoid , soft mass
- DDX : Small round cell tumors - lymphoma , embryonal rhabdomyosarcoma , Ewings sarcoma ..
- Metastasize widely ,
- surgery , radio & chemotherapy yield 5 year survival of 50 - 70 %.

• Larynx

Inflammations - most common disorders

- Laryngitis - allergic , viral , bacterial , chemical
 - more commonly as part of non specific URTI / tobacco smoke
 - part of systemic infection as in TB , diphtheria
 - most self limited , serious in infants & children : edema, congestion , exudation may cause obstruction

- Laryngoepiglottitis

- Caused by H.influenzae or streptococci
- Sudden swelling of epiglottis and vocal cords → obstruction
- uncommon in adult : large size of larynx and stronger accessory respiratory muscles

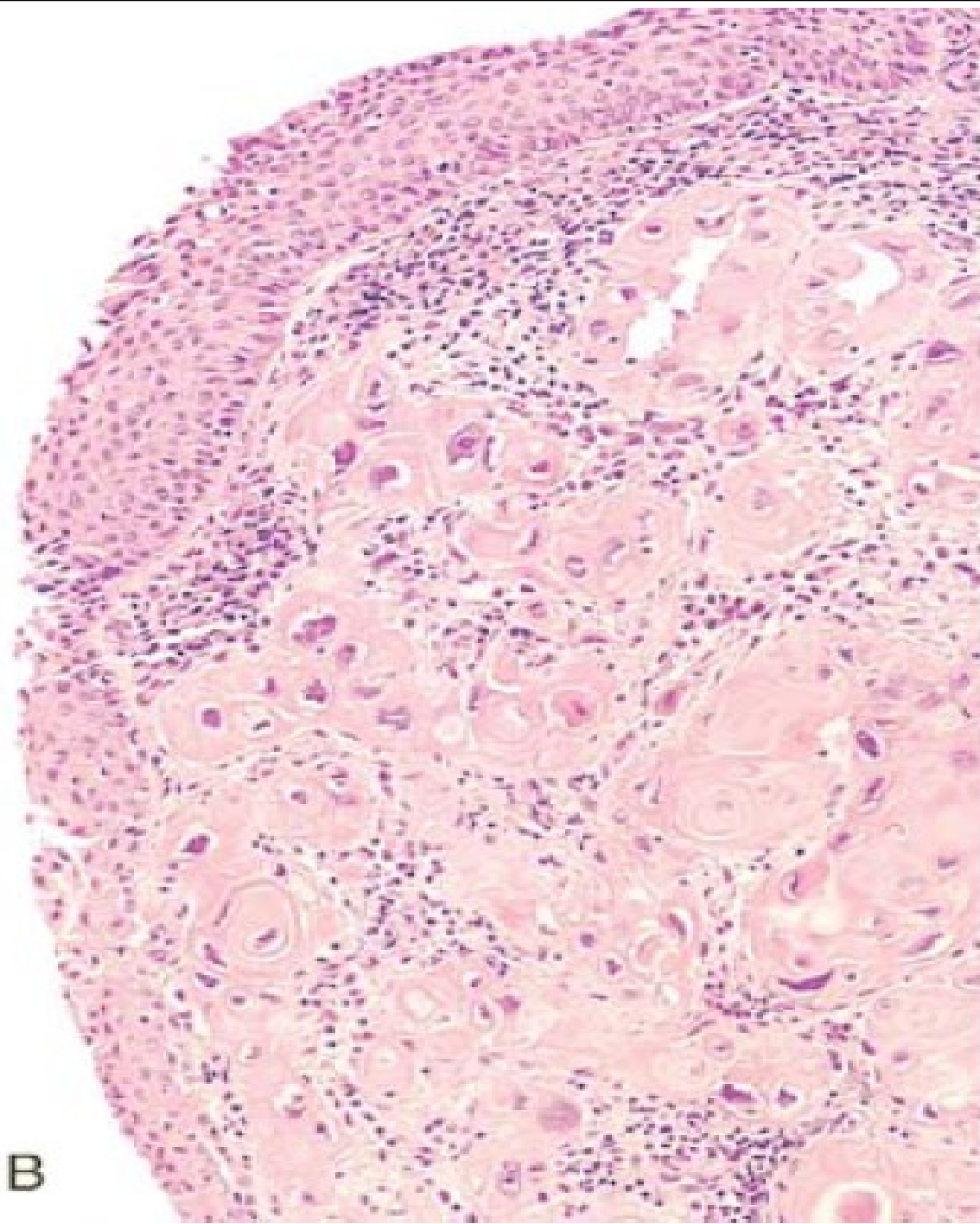
- Croup

- Laryngotracheobronchitis in children
- Narrowing of airways produces inspiratory stridor
- * laryngitis in heavy smokers predisposes to development of squamous epithelial changes

- (Vocal cord) Reactive nodules and polyps
 - Reactive , In heavy smokers and singers
 - Adult men
 - Smooth, rounded, sessile or pedunculated , small on true vocal cord
 - Progressive hoarseness , never give rise to carcinomas
- Carcinoma of larynx
 - 95 % Squamous cell carcinoma
 - Tobacco smoke , alcohol , nutritional factors, exposure to asbestos , irradiation , HPV
 - Manifested by persistent hoarseness



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Pulmonary infections

- Result when pulmonary defense mechanisms are impaired / when the resistance of the host in general is impaired
- Pulmonary defense mechanisms (clearing mechanism) can be impaired by
 - Loss or suppression of cough reflex
 - Coma, anesthesia, neuromuscular disorders, drugs, chest pain
 - May lead to aspiration of gastric contents

- Injury to mucociliary apparatus
 - Cigarette smoking, viral diseases, genetic disturbances
- Interference of action of alveolar macrophages
 - Alcohol, tobacco, anoxia
- Pulmonary congestion and edema
- Accumulation of secretions - bronchial obstruction

Pneumonia

- Any infection of the lung parenchyma
- One type of pneumonia predisposes to other
 - : Viral influenza epidemics - bacterial pneumonia
- Hematogenous seeding in to the lung difficult to differentiate from primary pneumonia
- Patients with chronic diseases acquire pneumonia while hospitalized

classification :

- Etiologic agent or clinical setting (if no etiologic agent identified)
- Pneumonia can arise in seven distinct clinical settings (Pneumonia syndromes)
 - Community acquired acute pneumonia
 - Community acquired atypical pneumonia
 - Nosocomial pneumonia
 - Aspiration pneumonia
 - Chronic pneumonia
 - Necrotizing pneumonia and lung abscess
 - Pneumonia in the immunocompromised host

Community acquired acute pneumonia

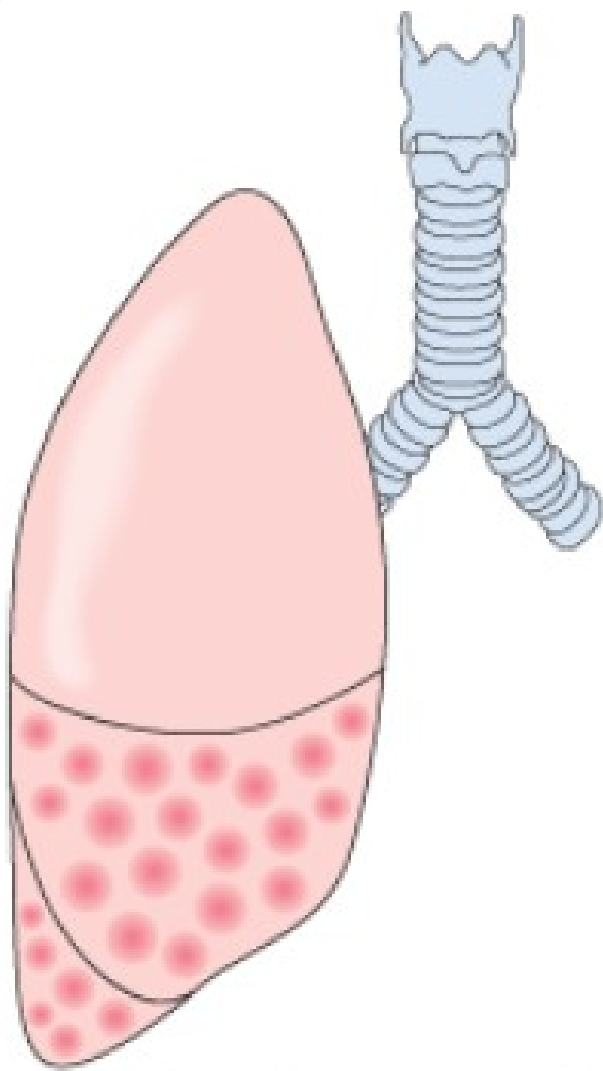
- Can be bacterial or viral
- Bacterial infection often follows an upper respiratory tract viral infection
- Bacterial invasion of the lung parenchyma causes alveoli to be filled with an inflammatory exudate, causing consolidation ('solidification') of the lung tissue.

Causing organisms

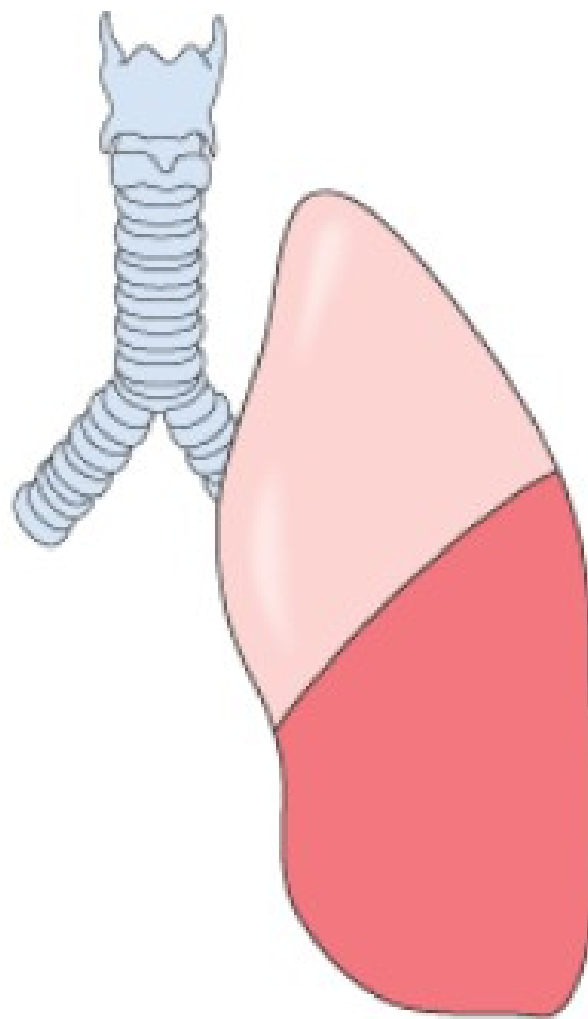
- **Streptococcus pneumoniae**
- **Hemophilus influenzae**
- **Staphylococcus aureus**
- **Klebsiella pneumoniae**

• Morphology

- Bacterial pneumonia has two gross patterns of anatomic distribution
 - Bronchopneumoia
 - Lobar pneumonia
- Patchy consolidation of the lung is characteristic of bronchopneumonia
- Lobar pneumonia results in fibrinosuppurative consolidation of large portion of a lobe or of an entire lobe
- The patchy involvement may become confluent, producing total lobar consolidation
- * causative agent & extent of disease - important clinically.



Bronchopneumonia



Lobar pneumonia



- In lobar pneumonia, four stages of the inflammatory responses have been described
 - Congestion
 - Red hepatization
 - Gray hepatization
 - Resolution
- Congestion
 - Lung is heavy, red
 - Vascular engorgement
 - Intra alveolar fluid with few neutrophils
 - Presence of numerous bacteria

- Red hepatization

- Massive confluent exudation with red cells, neutrophils, and fibrin filling alveolar spaces
- Grossly, the lobe appears as red, firm, and airless, with a liver like consistency - hepatization.

- Gray hepatization

- Follows with progressive disintegration of red cells and the persistence of a fibrinosuppurative exudate
- Grayish brown dry surface

- Resolution

- The consolidated exudate within the alveolar spaces undergoes progressive enzymatic digestion to produce a granular, semifluid, debris that is resorbed, ingested by macrophages, coughed up, or organized by fibroblasts
- Consolidation may extend to involve the pleura which may resolve leaving fibrous thickening or permanent adhesion

Bronchopneumonia :

- focal areas of consolidation , often bilateral , multilobar and basal.

- 3 - 4 cm in diameter, dry , granular
- Suppurative exudates fills bronchi, bronchioles, and adjacent alveolar spaces
- Complications of pneumonia
 - *Abscess formation* : tissue destruction & necrosis
 - *Empyema* : intra pleural fibrinosuppurative rxn
 - *Organization* - may convert a portion of the lung into solid tissues
 - *Bacteremic dissemination* -causing metastatic abscesses

- **Clinical course**
 - **Abrupt onset**
 - **Fever**
 - **Chills, productive cough, occasionally hemoptysis**
 - **Pleuritic pain and pleural friction rub when pleuritis**
 - **Radiology**
 - **Lobar pneumonia- radioopacity, usually well circumscribed**
 - **Bronchopneumonia shows focal opacities**

Community acquired atypical pneumonia

- Characterized by patchy inflammatory changes in the lungs, largely confined to the alveolar septa and pulmonary interstitium
- Atypical - denotes moderate amount of sputum, no physical findings of consolidation, only moderate elevation of white cell count, and lack of alveolar exudate
- Causes are mycoplasma , viruses, chlamydia

- Morphology
 - All agents - similar morphologic pattern
 - Red-blue , congested areas
 - Pleuritis or pleural effusions are uncommon
 - Predominant is the interstitial nature of the inflammatory reaction, virtually localized within the walls of the alveoli
 - Alveolar septa widened and edematous, infiltrated by lymphocytes, histiocytes, plasma cells
 - Alveoli may be free of exudate

Nosocomial pneumonia

- acquired in the course of a hospital stay
- Common : severe underlying disease, prolonged antibiotic therapy, invasive access devices , immunosuppression.
- Mechanical ventilators : higher risk
- Life threatening
- Gram negative rods and S.aureus

- Aspiration pneumonia

- Markedly debilitated patients or who aspirate gastric contents while unconscious or during repeated vomiting
- Chemical and bacterial cause , multi bacterial. Food also has irritative effect
- Often necrotizing, pursues a fulminant clinical course
- Frequent cause of death
- In those who survive, lung abscess formation is a common complication

- Lung abscess

- local suppurative process with in the lung, characterized by necrosis of lung tissue

- Etiology and pathogenesis

- Causative organisms are introduced by the following mechanisms
 - Aspiration of infective material
 - Antecedent primary bacterial infection
 - Septic embolism
 - Neoplasia
 - Direct trauma,
 - Spread of infections from a neighboring organ, hematogenous seeding

• Morphology

- Variable in diameter , any part , single or multiple
- Aspiration- more common on the right (because of the more vertical right main bronchus) and single
- Pneumonia/ bronchiectasis - mult, diffuse & basal
- Communication with airway - partial drainage of abscess - air containing cavity
- The cardinal histologic change in all abscesses is suppurative destruction of the lung parenchyma within the central area of cavitation

• Clinical course

- Cough, fever, foul smelling purulent sputum, chest pain, weight loss
- Dx : clinical + x - ray
- Always rule out underlying carcinoma (10 - 15%)
- Antibiotic treatment results in resolution
- Complications : empyema , hemorrhage , septic emboli , amyloidosis

- Chronic pneumonia

- Localized lesion in the immuno-competent patient
- Granulomatous inflammation due to bacteria or fungus
- Mtbcs , histoplasmosis , blastomycosis, coccidioidomycosis

- Pneumonia in the immunocompromised host

- Often , More than one agent involved
- Bacteria- *P. aeruginosa*, mycobacterium
- Viruses - CMV, Herpes virus
- Fungi - *P. carinii*, candida, aspergillus

TABLE 15-9 -- Causes of Pulmonary Infiltrates in Immunocompromised Hosts

Diffuse Infiltrate	Focal Infiltrate
<i>Common</i>	<i>Common</i>
Cytomegalovirus	Gram-negative rods
<i>Pneumocystis carinii</i>	<i>Staphylococcus aureus</i>
Drug reaction	<i>Aspergillus</i>
	<i>Candida</i>
	Malignancy
<i>Uncommon</i>	<i>Uncommon</i>
Bacteria	<i>Cryptococcus</i>
<i>Aspergillus</i>	<i>Mucor</i>
<i>Cryptococcus</i>	<i>Pneumocystis carinii</i>
Malignancy	<i>Legionella pneumophila</i>

Atelectasis

- Incomplete expansion of the lungs(neonatal) or to the collapse of previously inflated lung substance
- Airless pulmonary parenchyma
 - Reduces oxygenation and prediposes to infection
- Acquired , principally adults , divided in to three
 - Obstructive (absorptive)
 - Complete obstruction of airway
 - leads to absorption of oxygen trapped in the alveoli
 - *mediastinum may shift toward the atelectatic lung (diminished lung volume)*
 - Excessive secretions(mucus plungs) or exudates
 - Found commonly in asthma, chronic bronchitis, bronchiectasis , post OP, FB aspiration.

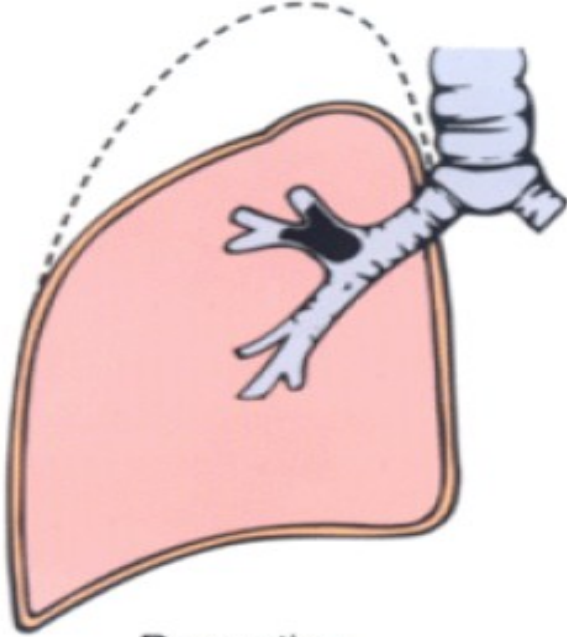
- **Compressive**

- When pleural cavity filled by fluid, tumor, air, blood
- Abnormal elevations of the diaphragm
- Mediastinum shifts away from the affected lung

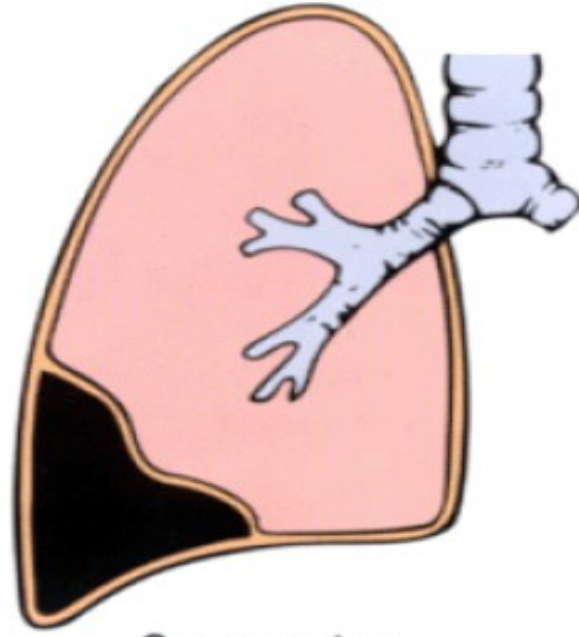
- **Contraction**

- Develops when generalized or local fibrotic changes in the lung or pleura prevent full expansion

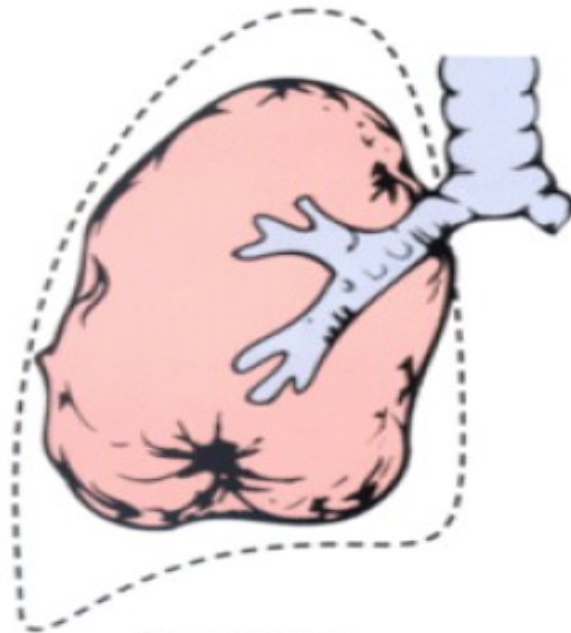
- Atelectasis is a reversible disorder (except that caused by contraction)



Resorption



Compression



Contraction

Pulmonary Tuberculosis

• Primary Pulmonary Tuberculosis

- Lungs are the usual site of primary infection
- Ghon complex- initial focus of primary infection
 - consists of Parenchymal subpleural lesion,
Enlarged caseous lymph nodes (bronchial + hilar)
 - In most cases, patients are asymptomatic
 - Lesions undergo fibrosis and calcification
 - In immunodeficient adults or children, progressive spread may follow
 - Progressive Primary tuberculous pneumonia
 - Miliary disseminated tuberculosis

• Secondary (Reactivation) Pulmonary Tuberculosis

- Located in the apex of the lungs, less commonly in the hilus
- Begins as small focus of consolidation
- Histologically - coalescent granulomas
- May heal spontaneously or with therapy or may progress

• Progressive Pulmonary Tuberculosis (from secondary tuberculosis)

- Active lesions progress to cause further pulmonary and distant organ involvement
 - Cavitory fibrocaceous tuberculosis
 - Miliary tuberculosis
 - Tuberculous bronchopneumonia

- Cavitary fibrocaceous tuberculosis
- Erosion in to bronchiole with drainage of caseous focus
- Growth of tubercle bacilli is favored - oxygen
- Infection may disseminate through airways, via lymphatics, blood
- Pleural effusions, endobronchial and endotracheal tuberculosis
- Laryngeal seeding and intestinal tuberculosis

- **Miliary tuberculosis**

- Result of lymphohematogenous dissemination
- Confined only to the lungs or involving other organs (spleen , liver , BM ,kidneys)
- Tuberculous infection may drain via lymphatics into the right side of the heart, then to the lungs
- Some bacilli pass to the systemic circulation and create distant seedings
- Miliary lesions are distinct multiple , yellow-white, firm areas of consolidation

- **Tuberculous bronchopneumonia**

- Tuberculous infection spread rapidly through out large areas of lung parenchyma
- Sometimes, well developed tubercles do not form
- Numerous bacilli usually present

Isolated organ tuberculosis

- When disseminated organisms establish infection in only one or two organs
- Adrenals , kidneys , bone ,FGT

Obstructive versus Restrictive Pulmonary Diseases

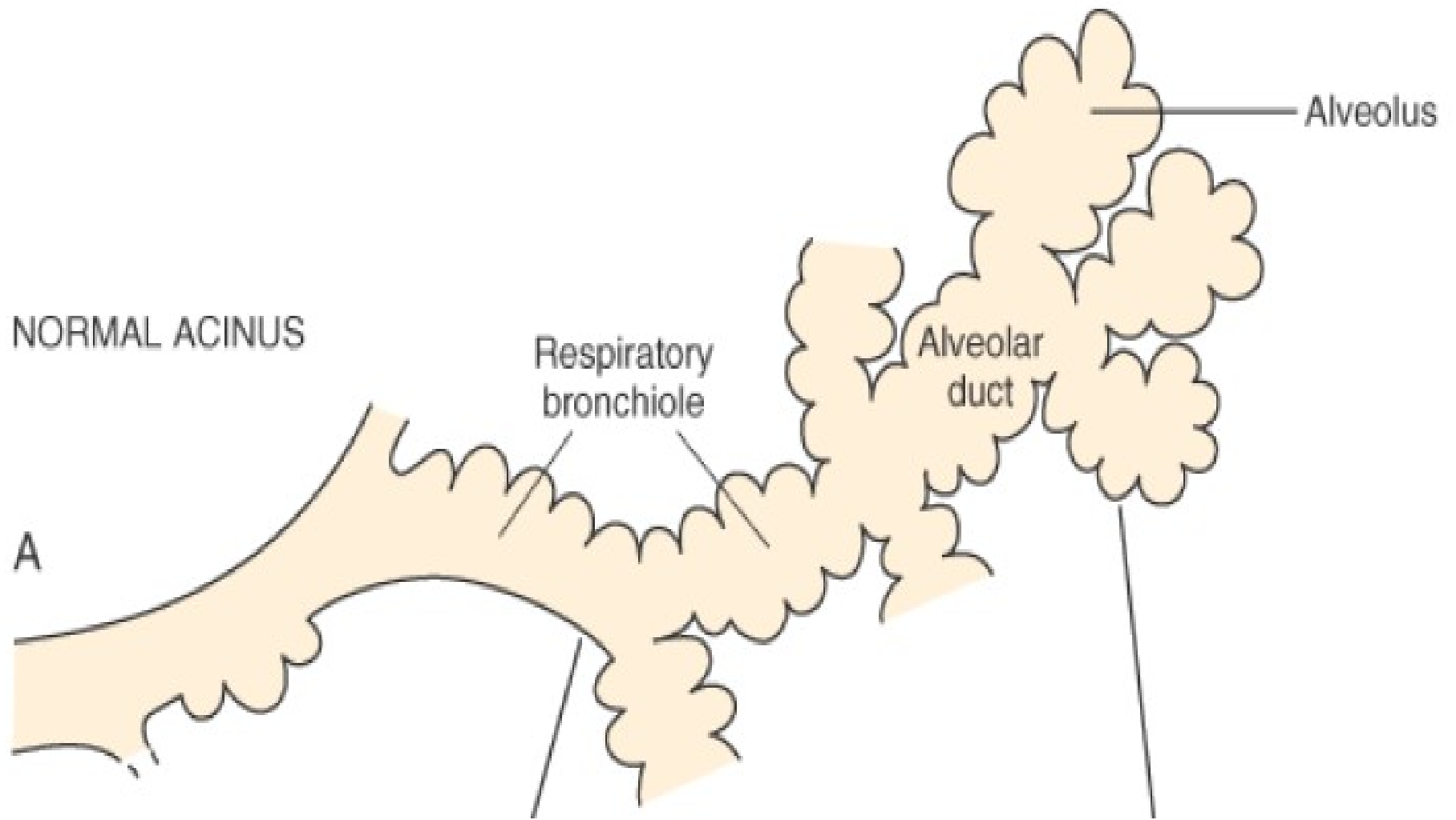
- Diffuse pulmonary disease
- Classification based on pulmonary function test

Restrictive diseases

- Characterized by reduced expansion of lung parenchyma = decreased total lung capacity
- Defects occur in two general conditions
 1. Chest wall disorders with normal lung (neuromuscular disorders PM, severe obesity, pleural diseases,)
 2. Acute or chronic interstitial and infiltrative diseases

- Obstructive disease (Airway disease)
- Chronic obstructive pulmonary diseases(COPD)
 - Characterized by an **increase in resistance to airflow** owing to partial or complete obstruction at any level(trachea to alveoli)
 - Major diffuse obstructive disorders are
 - **Emphysema**
 - **Chronic bronchitis**
 - **Bronchiectasis**
 - **Asthma**

- Obstructive pulmonary diseases
 - Share a major symptom- dyspnea - and are accompanied by chronic or recurrent obstruction to airflow with in the lung
 - Emphysema
 - Characterized by abnormal permanent enlargement of the airspaces distal to the terminal bronchiole, accompanied by destruction of their walls and without obvious fibrosis
 - The enlargement of airspaces unaccompanied by destruction is termed Over inflation



- Types of emphysema : based on anatomic distribution with in lobule
- Lobule : cluster of acini (alveolated terminal respiratory units)
- The four major types :
 1. Centriacinar
 2. Panacinar
 3. Paraseptal
 4. Irregular
 - * Centri and panacinar cause clinically significant obstruction.

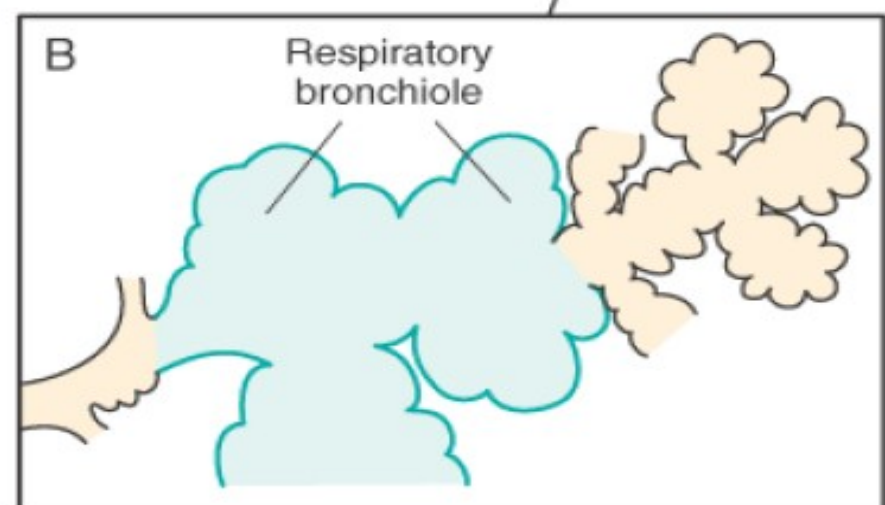
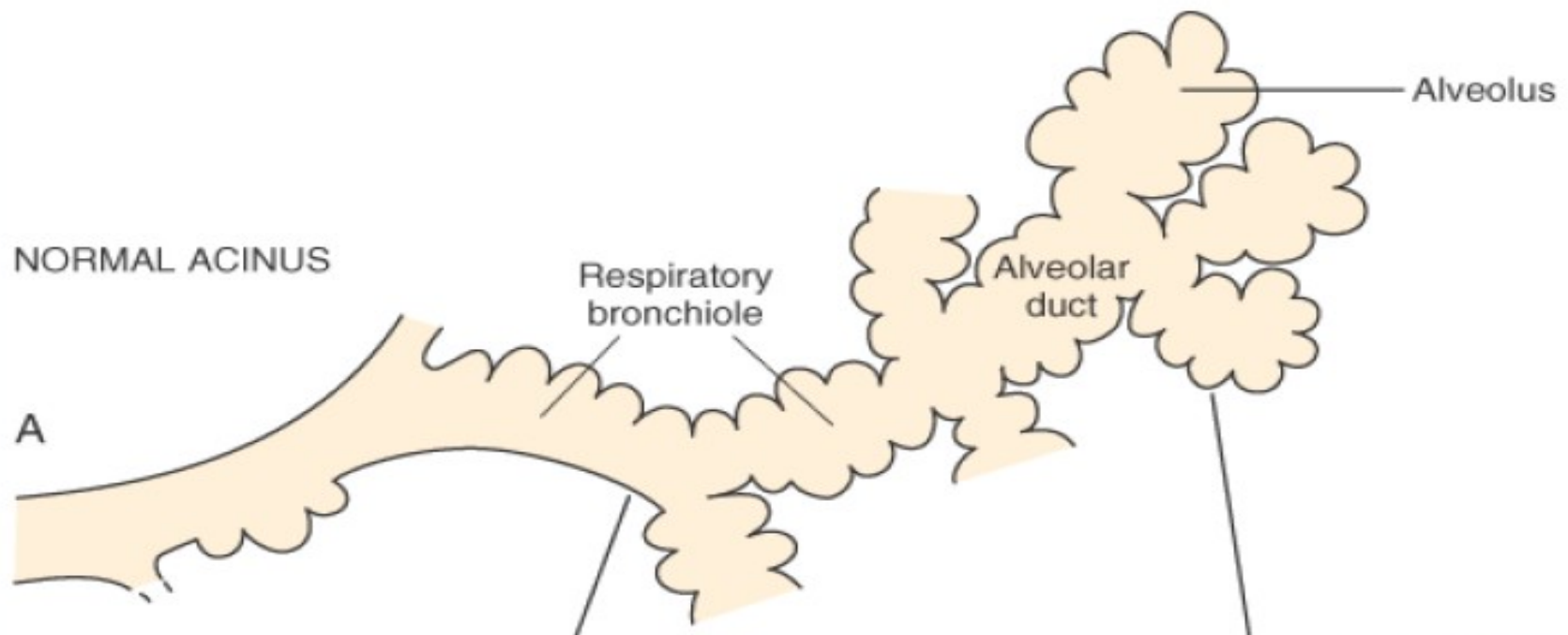
◆ Centriacinar (Centrilobular)

- The central or proximal parts of the acini, formed by respiratory bronchioles are affected, where as distal alveoli are spared
- 95 % of cases of emphysema
- More common and severe in the upper lobes
- Occurs predominantly in **heavy smokers**

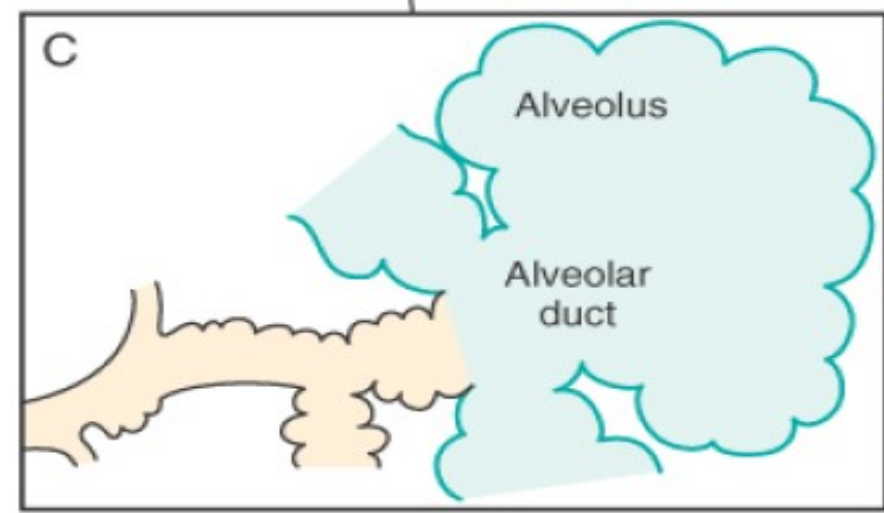
◦ Panacinar (Panlobular) Emphysema

- Acini are uniformly enlarged from the level of the respiratory bronchiole to the terminal blind alveoli
- Tends to occur more in the lower zones
- It is associated with α 1-antitrypsin def.

- **Distal acinar (Paraseptal) Emphysema**
 - The proximal portion of the acinus is normal, but the distal part is predominantly involved
 - More striking adjacent to the pleura
 - Occurs adjacent to areas of fibrosis, scarring or atelectasis
 - More severe in the upper half of the lungs
 - Underlies many of the spontaneous pneumothorax



Centriacinar emphysema



Panacinar emphysema

- Airspace enlargement with fibrosis
(Irregular Emphysema)

- The acinus is irregularly involved
- Associated with scarring (healed inflammatory scar)

- Pathogenesis

- **Protease - antiprotease theory**

- Alveolar wall destruction occurs from an imbalance between proteases (mainly elastases) and antiproteases in the lungs
- Any stimulus that increases either the number of leukocytes in the lung or the release of their elastase- containing granules increases elastolytic activity
- Stimulated neutrophils also release oxygen free radicals which inhibit α 1-AT activity
- Hereditary deficiency of α 1-AT

- With low levels of serum α 1-AT, the process of elastic tissue destruction is unchecked, with consequent emphysema
- In smokers, both increased elastase availability and decreased antielastase activity occur
- In smokers, both neutrophils and macrophages accumulate in the alveoli
 - It may involve the direct chemo attractant effect of nicotine as well as reactive oxygen species contained in smoke
 - Accumulated neutrophils are activated and release their enzymes
 - Increases elastase activity of macrophages
 - Smoking also plays a role in perpetuating the oxidant- antioxidant imbalance

• Clinical course

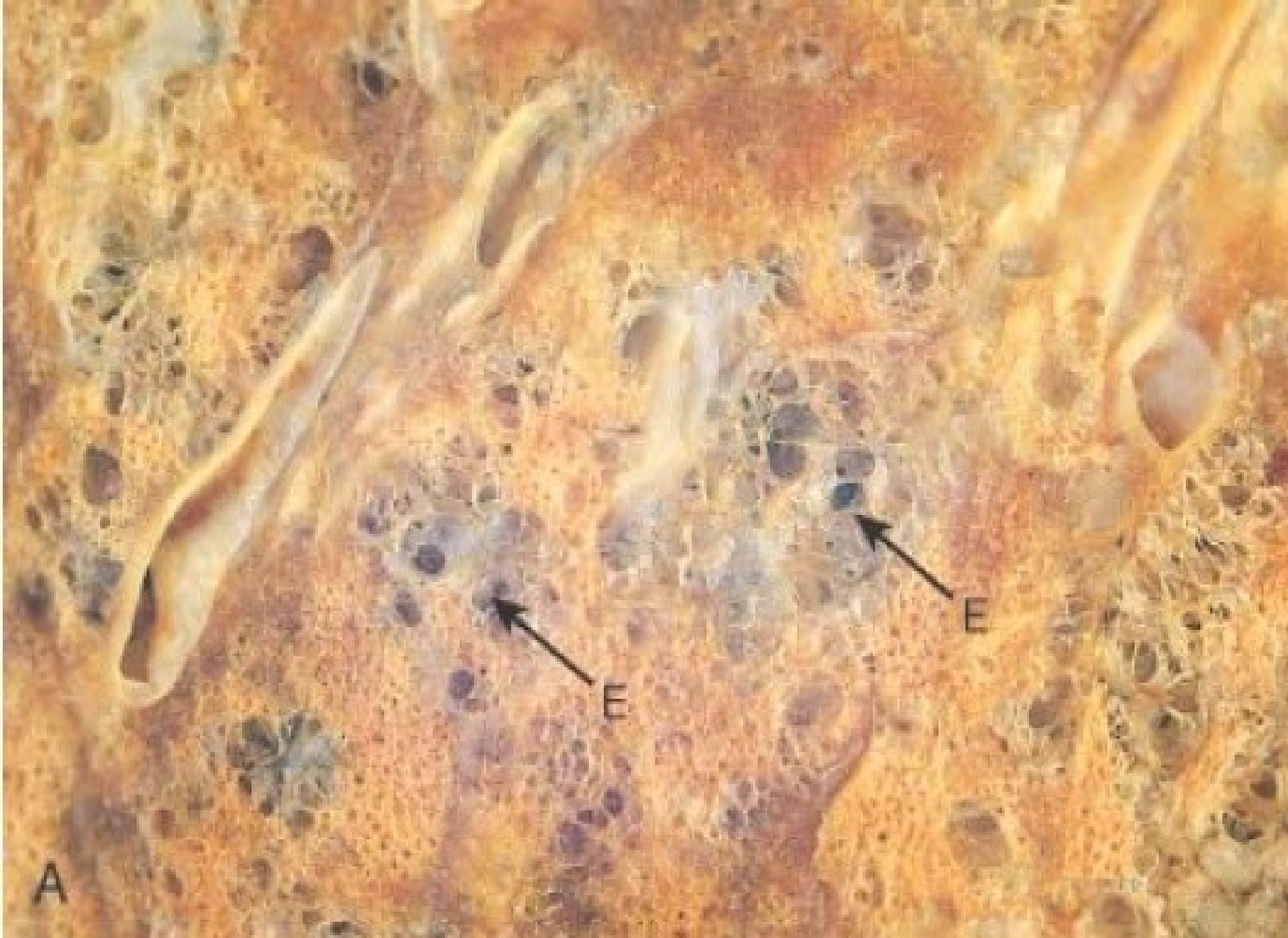
- Do not appear until one third of pulmonary parenchyma is damaged
- Dyspnea
- Cough
- Weight loss
- Expiratory airflow limitation by spirometry
- Emphysema patients- pink puffers
- Chronic bronchitis- blue bloaters
- Development of cor-pulmonale and congestive heart failure secondary to pulmonary hypertension

- **Morphology**

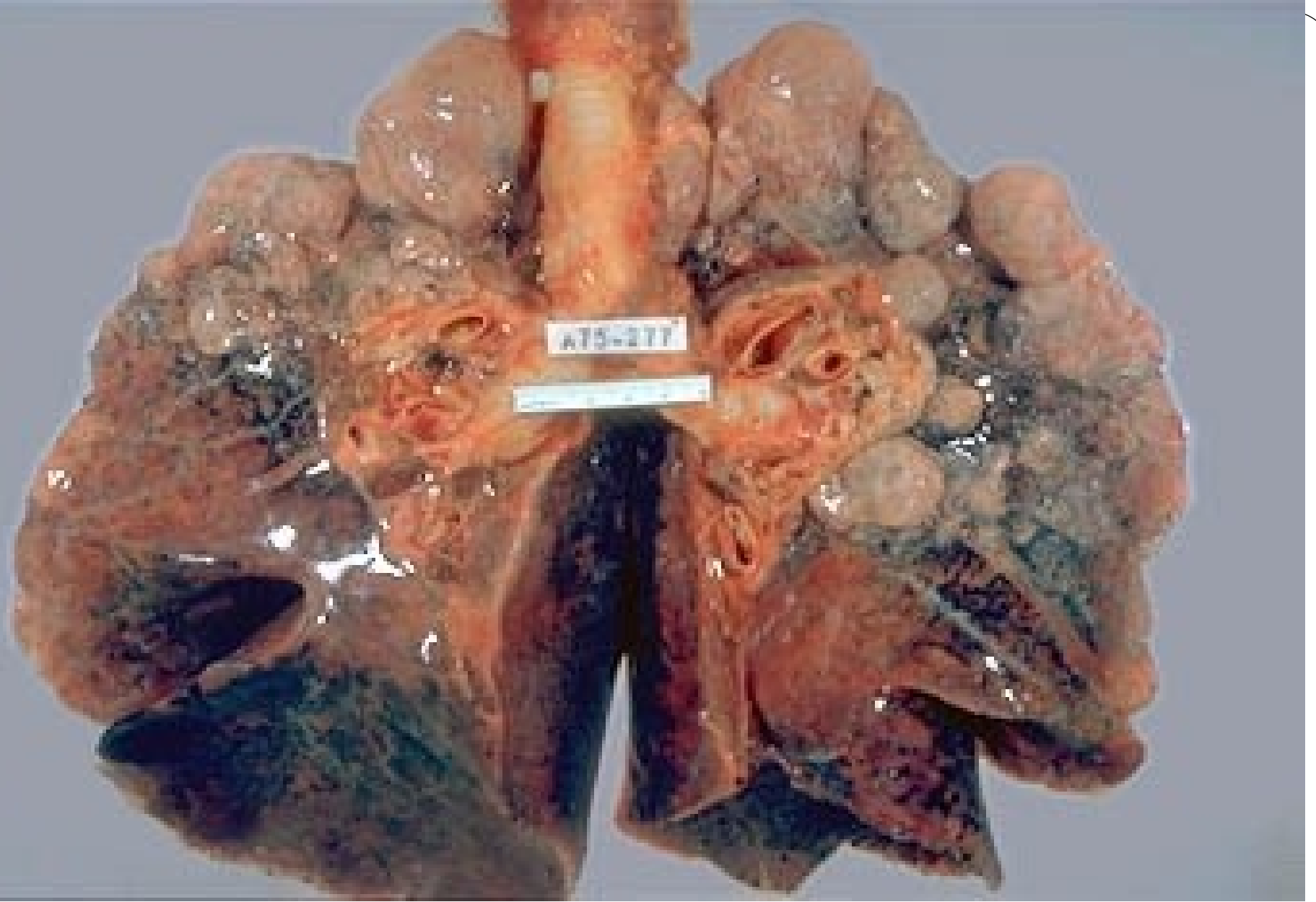
- Panacinar emphysema produces voluminous lungs
- Large apical blebs or bullae are characteristic of irregular emphysema and distal acinar emphysema
- Respiratory bronchioles and vasculature of the lung are deformed

- **Cause of death in COPD**

- Right sided heart failure
- Massive collapse of lungs secondary to pneumothorax



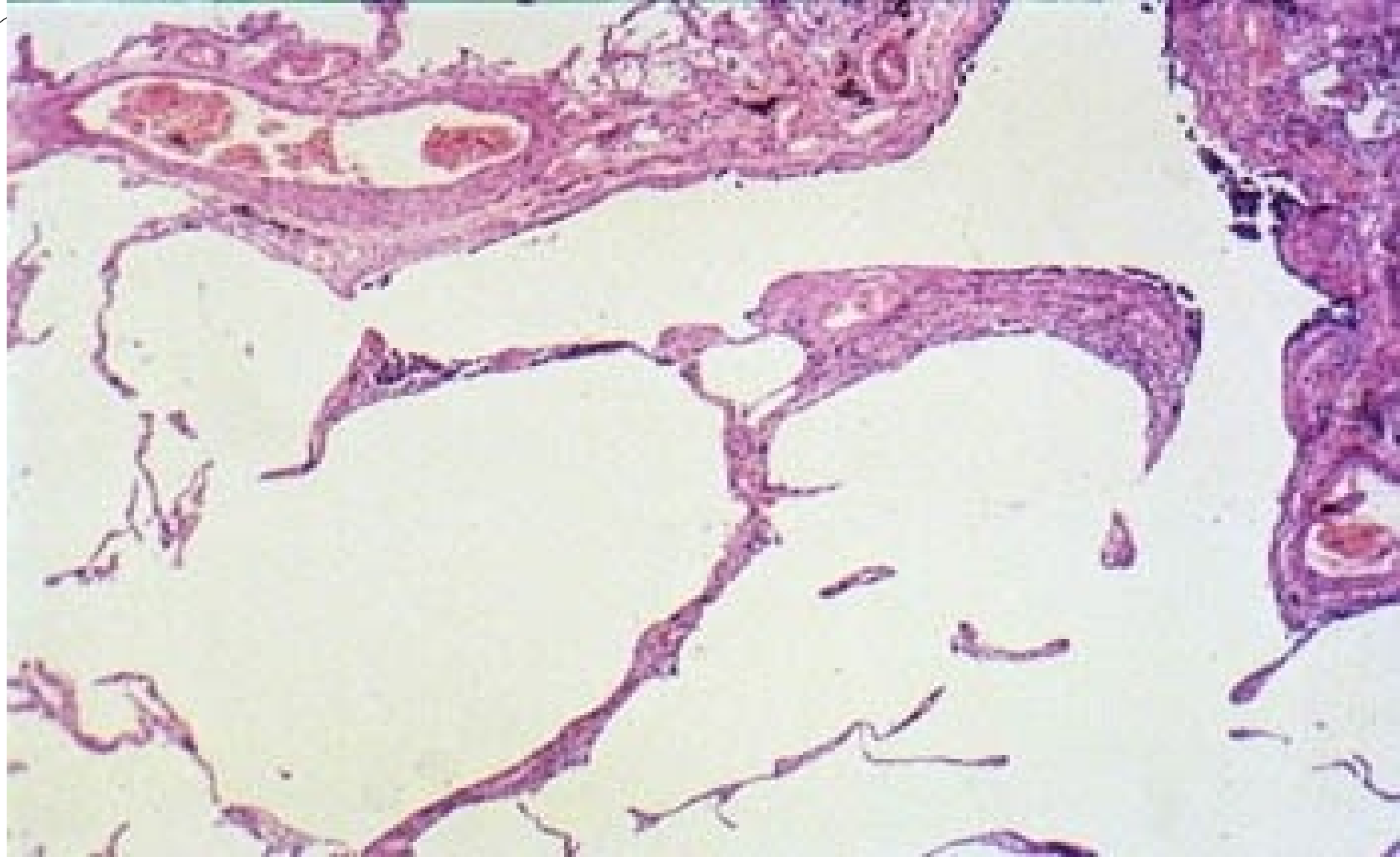




- Lungs, emphysema - Gross



- Lung, centrilobular emphysema - Gross, cut surface



- Lung, emphysema - **enlarged , thin walled air spaces with compression of septal capillaries = bloodless**

• Chronic Bronchitis

- Clinical def .: Persistent cough with sputum production for at least 3 months in at least 2 consecutive years, in the absence of any other identifiable cause

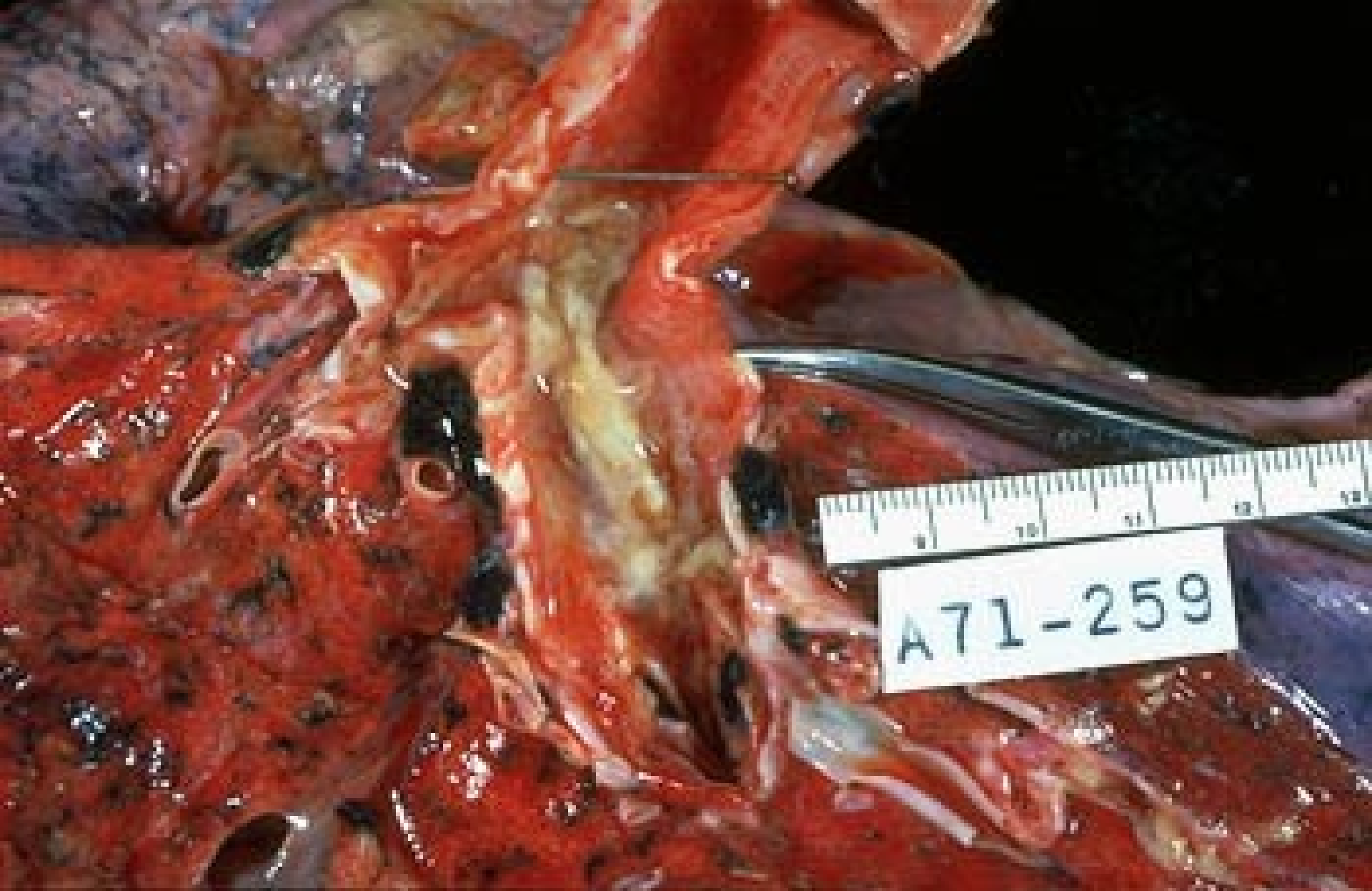
• Pathogenesis

- Chronic irritation by inhaled substances(esp . Tobacco smoke)
- Bacterial and viral infections are important in triggering acute exacerbation

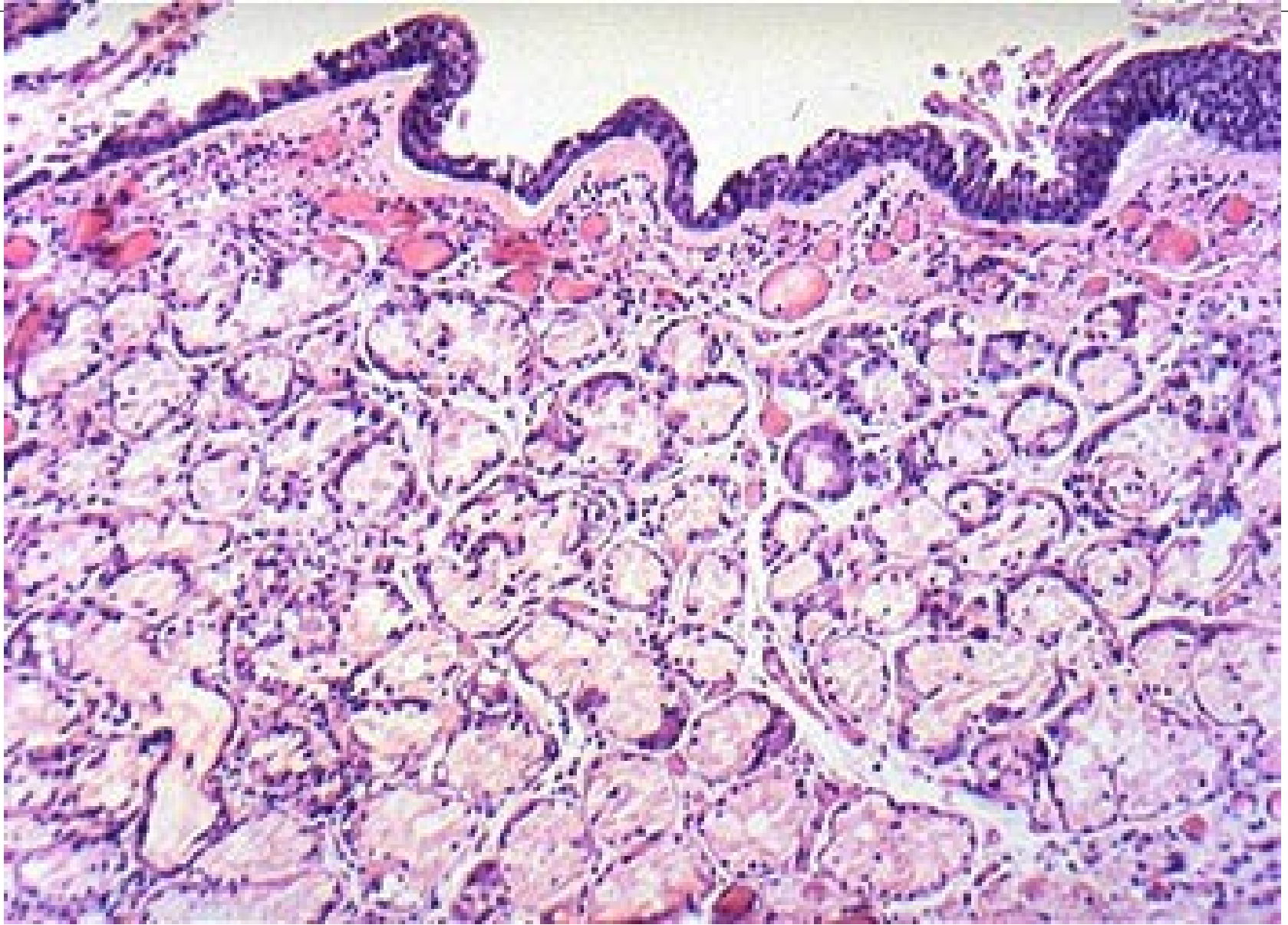
- Most frequent in middle-aged men
- More common in heavy smokers
- Hyper secretion of mucus in the large airways, associated with hypertrophy of the submucosal glands
- Proteases from neutrophils and matrix metalloproteinases stimulate this hypersecretion
- Role of infection appears to be secondary

- Morphology

- Hyperemia, swelling, edema of mucus membranes accompanied by excessive secretion
- Chronic inflammation of the airways, and enlargement of mucus secreting glands
- Bronchial epithelium may exhibit squamous metaplasia and dysplasia
- Marked narrowing of bronchioles



- Lung, mucopurulent bronchitis - Gross, cut surface



- Lung, chronic bronchitis

- **Clinical features**
 - **Persistent productive cough**
 - **Dyspnea**
 - **COPD**
 - **Cor pulmonale with cardiac failure**

	Predominant Bronchitis	Predominant Emphysema
Age (yrs)	40-45	50-75
Dyspnea	Mild; late	Severe; early
Cough	Early; copious sputum	Late; scanty sputum
Infections	Common	Occasional
Respiratory insufficiency	Repeated	Terminal
Cor pulmonale	Common	Rare; terminal
Airway resistance	Increased	Normal or slightly increased
Elastic recoil	Normal	Low
Chest radiograph	Prominent vessels; large heart	Hyperinflation; small heart
Appearance	Blue bloater	Pink puffer

◦ Bronchial Asthma

- Chronic relapsing inflammatory disorder featuring increased responsiveness of the tracheo bronchial tree to various stimuli , resulting in a paroxysmal contraction of airways
- Wheezing, breathlessness, chest tightness, and cough, particularly at night and/or in the early morning
- wide spread broncho constriction and airflow limitation that is partly reversible , spontaneously or with treatment.
- Eosinophils, mast cells, macrophages, T-lymphocytes, neutrophils, and epithelial cells play a role

- Pathogenesis

- Major etiologic factors
- Genetic predisposition to type I hypersensitivity,
- Acute and chronic airways inflammation,
- Bronchial hyper responsiveness

- Atopic (Allergic) asthma

- Usually begins in childhood , most common
 - Triggered by environmental antigenes (dust , pollen, animal dander, foods)
 - Positive family history of atopy is common
 - Asthmatic attacks often preceded by allergic rhinitis, urticaria, or eczema
 - Classic example of type I IgE mediated hypersensitivity reaction.
-
- Initial sensitization to inhaled antigens which stimulate induction of T_H2 type cells that release cytokines such as IL-4 and IL-5

- These cytokines promote IgE production by B cells, growth of mast cells, and growth and activation of eosinophils
- Subsequent IgE mediated reaction to inhaled allergens elicits an acute response and a late phase reaction
- Acute or immediate response consists of bronchoconstriction, edema, mucus secretion
- Exposure of presensitized IgE coated mast cells to antigen stimulates release of chemical mediators

- Mast cells also release cytokines that cause the influx of other leukocytes and set the stage of the Late phase reaction
- Mediators can also be produced by other cells
 - Inflammatory cells already present in asthmatics
 - Vascular endothelium
 - Airway epithelial cells

- Non atopic asthma

- Most frequently triggered by respiratory infection
- No evidence of IgE mediated hypersensitivity
- Positive family history uncommon
- Serum IgE levels are normal
- Negative skin test
- No associated allergies
- Inhaled air pollutants may also contribute
- The primary cause of increased air way reactivity is unknown

- Drug induced asthma

- Several pharmacological agents
- Aspirin - sensitive asthma : uncommon , small dose.
- elaborate bronchoconstrictor leukotrienes

- Occupational asthma

- Fumes (epoxy resins, plastics) , organic & chemical dusts (wood, cotton, platinum), gases (toluene) , formaldehyde, penicilline products
- Small amount triggers , after repeated exposure.
- Type I and other hypersensitivity, direct liberation of bronchconstrictor substances , hypersensitivity response of unknown origin.

- Morphology

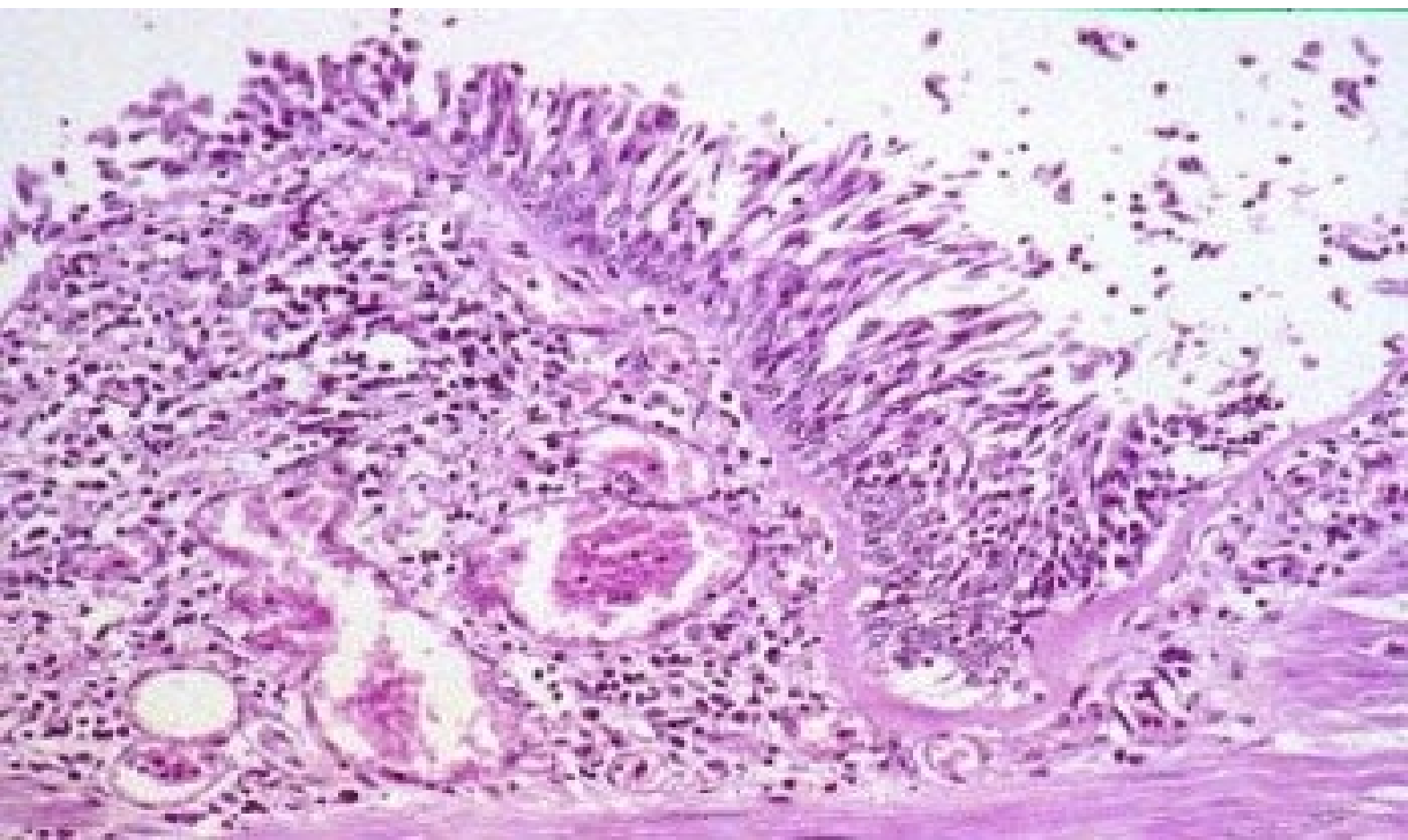
- Gross

- Lungs :over inflated , patchy atelectasis with
- Occlusion of bronchi and bronchioles by mucus plugs

- Microscopy:

- Airway remodelling

- Thickening of the basement membrane of the bronchial epithelium
- Edema and inflammatory infiltrate in the bronchial walls, with prominence of eosinophils and mast cells
- Increase in size of the submucosal glands
- Hypertrophy of the bronchial muscle wall



- Bronchus, bronchial asthma
- Airflow obstruction – primarily muscular broncho constriction; remodeling may contribute.

- Bronchiectasis

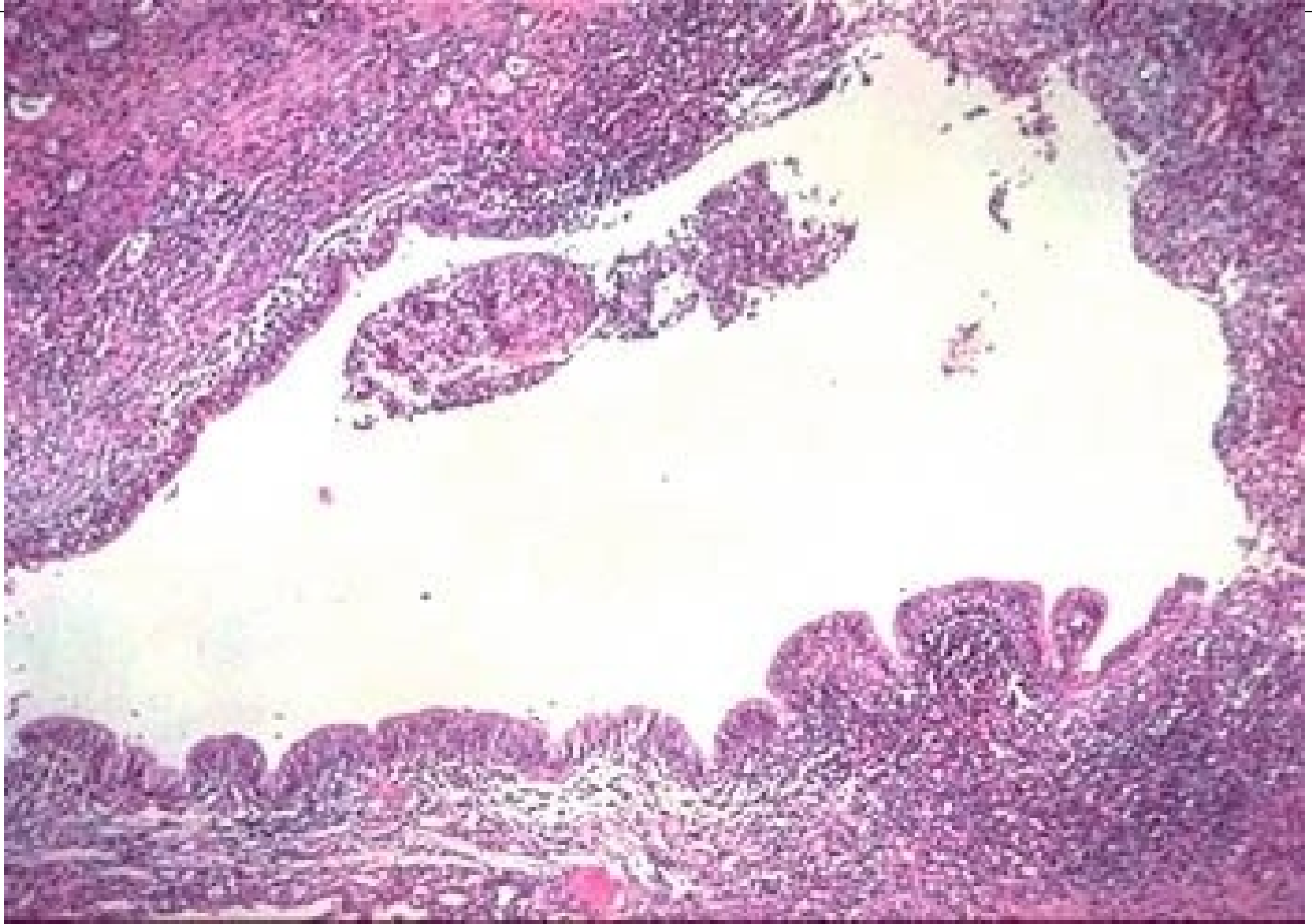
- Characterized by permanent dilation of bronchi and bronchioles caused by destruction of the muscle and elastic tissue, resulting from or associated with chronic necrotizing infections
- Manifested by cough, fever, and copious amounts of foul smelling, purulent sputum
- Bronchiectasis develops in association with
 - Congenital or hereditary conditions including cystic fibrosis , immunodeficiency states
 - Post infectious conditions (bacterial, viral , fungal)

- Bronchial obstruction (tumor , FB , mucus plugging)
- Other conditions : RA, SLE, IBD, post transplantation
 - Etiology and pathogenesis
- Obstruction and infection are major influences
- Pooling of secretions distal to obstruction, and there is inflammation of the airway
- Severe infections of the airway lead to inflammation, often with necrosis, fibrosis, and eventually dilation of the airways
 - Morphology
- Usually affects the lower lobes bilaterally
- More severe in the more distal bronchi and bronchioles

- The airways are dilated (sts up to 4X)
- Cut surface of the lung reveals dilated bronchi appearing as cysts filled with mucopurulent secretions
- Histology : varies
- Active case- intense acute and chronic inflammatory exudation associated with desquamation of lining epithelium and areas of ulceration
- Chronic cases- fibrosis and obliteration of lumina



- Lung, bronchiectasis



- Bronchiole, bronchiectasis

- **Clinical course**
 - **Severe, persistent cough**
 - **Expectoration of foul smelling, sometimes bloody sputum**
 - **Dyspnea and orthopnea**
 - **Fever**
 - **Cor pulmonale**
 - **Metastatic brain abscesses**
 - **amyloidosis**

TABLE 15-3 -- Disorders Associated with Airflow Obstruction: The Spectrum of Chronic Obstructive Pulmonary Disease

Clinical Term	Anatomic Site	Major Pathologic Changes	Etiology	Signs/Symptoms
Chronic bronchitis	Bronchus	Mucous gland hyperplasia, hypersecretion	Tobacco smoke, air pollutants	Cough, sputum production
Bronchiectasis	Bronchus	Airway dilation and scarring	Persistent or severe infections	Cough, purulent sputum, fever
Asthma	Bronchus	Smooth muscle hyperplasia, excess mucus, inflammation	Immunologic or undefined causes	Episodic wheezing, cough, dyspnea
Emphysema	Acinus	Airspace enlargement; wall destruction	Tobacco smoke	Dyspnea

- Restrictive disease
 - Characterized by reduced expansion of lung parenchyma, with decreased total lung capacity
- Restrictive diseases occur in
 - Chest wall disorders
 - Acute or chronic interstitial and infiltrative diseases

• Diffuse Interstitial (Infiltrative, Restrictive) diseases

- heterogeneous group of disorders characterized by diffuse and chronic involvement of the pulmonary connective tissue, principally the most peripheral and delicate interstitium in the alveolar walls
- Interstitial : BM of endothelial & epithelial cells, collagen fibers, elastic tissue , proteoglycans , fibroblasts , mast cells,...
- Many : unknown etiology & pathogenesis ;
- Some: intra alveolar & interstitial component ; frequent overlap in histologic features
- Similar SSx , radiologic findings & patho physiologic alteration

- Patients have dyspnea, tachypnea, end respiratory crackles, cyanosis, with out wheezing or other evidence of airway obstruction
- Eventually, pulmonary hypertension and right sided heart failure with cor pulmonale may result.

Similar clinical , radiologic and pathologic changes

- **Clinical: dysnea , decreased lung volume and compliance**
- **Radiologic: diffuse infiltrates**
- **Pathologic : diffuse , chronic inflammation or fibrosis of alveolar interstitium**

- Pneumoconioses

- Used to describe the non neoplastic lung reaction to inhalation of mineral dusts, organic & inorganic particulates, chemical fumes and vapors either in occupational exposure or particulate air pollution.

TABLE 15-6 -- Lung Diseases Caused by Air Pollutants

Agent	Disease	Exposure
<i>Mineral Dusts</i>		
Coal dust	Anthracosis	Coal mining (particularly hard coal)
	Macules	
	Progressive massive fibrosis	
	Caplan syndrome	
Silica	Silicosis	Foundry work, sandblasting, hardrock mining, stone cutting, others
	Caplan syndrome	
Asbestos	Asbestosis	Mining, milling, and fabrication; installation and removal of insulation
	Pleural plaques	
	Caplan syndrome	
	Mesothelioma	
	Carcinoma of the lung, larynx, stomach, colon	

Beryllium	Acute berylliosis	Mining, fabrication
	Beryllium granulomatosis	
	Bronchogenic carcinoma (?)	
Iron oxide	Siderosis	Welding
Barium sulfate	Baritosis	Mining
Tin oxide	Stannosis	Mining
<i>Organic Dusts That Induce Hypersensitivity Pneumonitis</i>		
Moldy hay	Farmer's lung	Farming
Bagasse	Bagassosis	Manufacturing wallboard, paper
Bird droppings	Bird-breeder's lung	Bird handling
<i>Organic Dusts That Induce Asthma</i>		
Cotton, flax, hemp	Byssinosis	Textile manufacturing
Red cedar dust	Asthma	Lumbering, carpentry
<i>Chemical Fumes and Vapors</i>		
Nitrous oxide, sulfur dioxide, ammonia, benzene, insecticides	Bronchitis, asthma	Occupational and accidental exposure
	Pulmonary edema	
	ARDS *	
	Mucosal injury	
	Fulminant poisoning	

- Coal workers pneumoconiosis

- Lung findings range from
 - Asymptomatic anthracosis
 - carbon pigment accumulate in the connective tissue along the lymphatics
 - Simple coal workers pneumoconioses
 - coal macules and nodules
 - Little / no lung dysfunction
 - Progressive massive fbrosis
 - intensely blackened scars larger than 2 cm
 - dense collagen and pigment
 - Impaired lung function

- Silicosis

- caused by inhalation of silica
- Slowly progressing, nodular, fibrosing pneumoconioses

- Asbestos related diseases

- Occupational exposure to asbestos is related to
 - Localized fibrous plaques
 - Pleural effusions
 - Parenchymal interstitial fibrosis (asbestosis)
 - Bronchogenic carcinoma
 - Mesotheliomas
 - Laryngeal and colon carcinomas

Sarcoidosis

- Systemic , unknown cause , non caseating granulomas
- Varied clinical presentation : bilateral hilar LAP or lung involvement on CXR 90% of cases
- Eye and skin lesions follow
- Histologic diagnosis is by exclusion
- Women more affected

Etiology & Pathogenesis

Unknown

Disordered immune regulation

Genetic predisposition

Environmental agents

- Immunologic abnormalities : intraalveolar & interstitial accumulation of CD4+ Th cells , increased levels of cytokines , anergy to common skin test antigens (candidia , PPD)
 - Genetic factors : familial and racial clustering of cases , association with certain HLA genotypes
 - Environmental factors : proposed include mycobacteria , P.acne , rickettsia species.
- * no evidence to suggest infectious agent as a cause.

Morphology

- All involved tissue : non caseating granuloma ,
Langehans or foreign body type giant cells
- Schaumann bodies : laminated concretions
composed of calcium and proteins
- Asteroid bodies : stellate inclusions within the
giant cells of granuloma in 60 % of cases
- Characteristic not pathognomonic
- Can involve any organ in the body

lungs

- Common site of involvement
- Gross: no lesion , small nodules
- Mic: along lymphatics, around bronchi, Bv
- Alveolar lesions can be seen
- Lesions tend to heal : varying stages of fibrosis and hyalinization
- Pleura: sometimes involved

lymphnodes

- In almost all cases
- Hilar and mediastinal (any node)
- Enlarged, discrete , calcified
- Tonsils : $\frac{1}{4}$ - $\frac{1}{3}$ of cases.
- Spleen , liver and bone marrow
- Skin , eye with associated glands , salivary glands , muscle are other favored organs

Clinical course

- Protean clinical disease
- Great majority of cases : pulmonary compliant and non specific symptoms
- Unpredictable course

• Lung Tumors

- Vast majority are carcinomas (90-95%)
- 5% bronchial carcinoids
- 2 – 5 % mesenchymal and miscellaneous

• Carcinomas

- Most 40 and 70 years , peak 5 & 6 decades
- Prognosis is bad
- Etiology and pathogenesis
 - Tobacco smoking
 - Industrial hazards
 - Radiation, asbestos , uranium
 - Air pollution

• Classification

- Squamous cell carcinoma
- Adenocarcinoma *
- Small cell carcinoma
- Large cell carcinoma

* For clinical use , the various histologic subtypes can be clustered in to two groups : likelihood of metastasis & response to therapies

Small cell carcinomas: often metastatic , high initial response to chemotherapy

Non small cell carcinomas : less often metastatic , less responsive

- **Morphology**

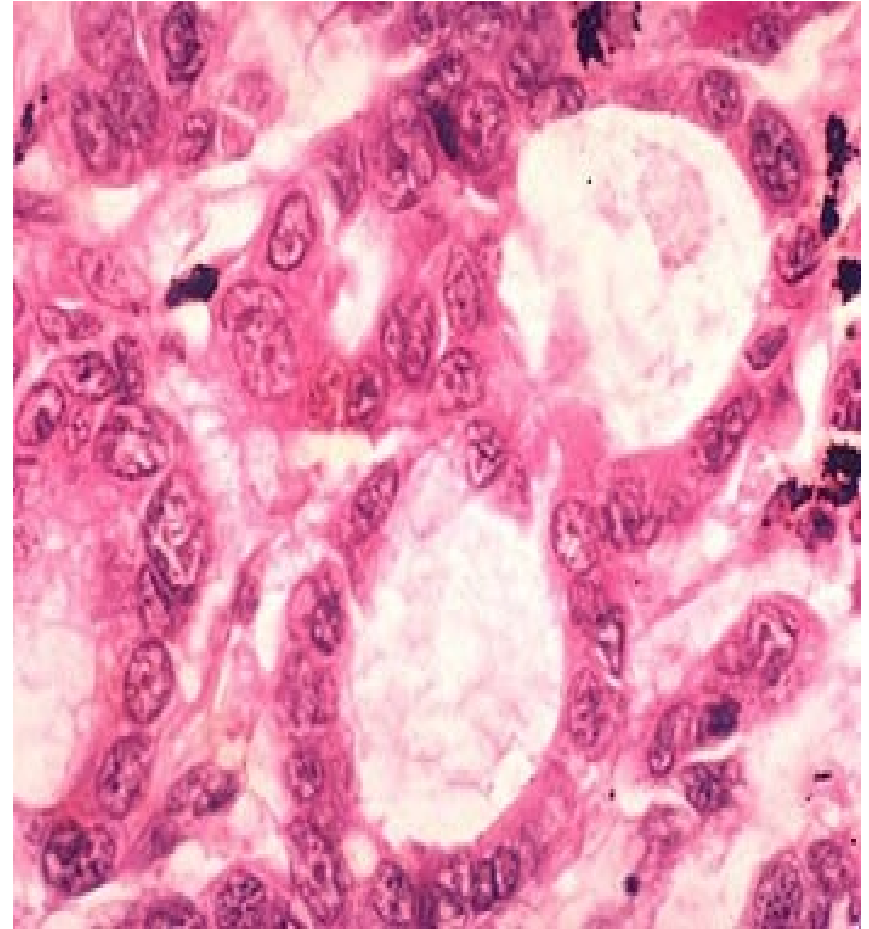
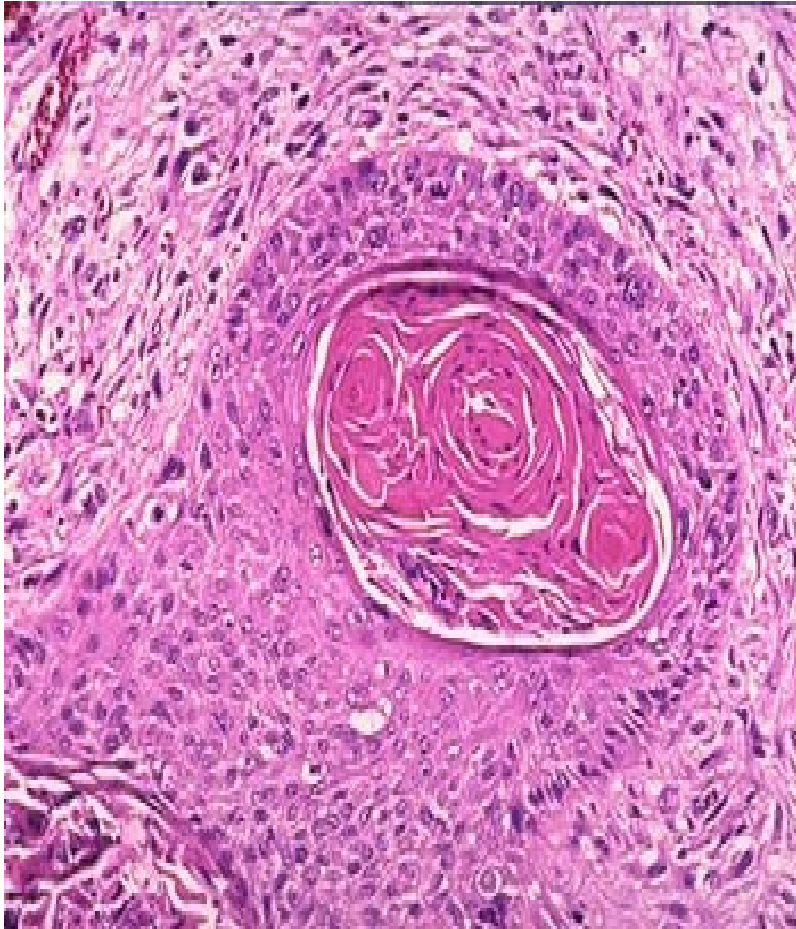
- Lung carcinomas arise most often **in and about the hilus of the lung**
- About **three fourth** take their origin from **bronchi**
- A small percentage from the peripheral lung tissue (alveolar septal cells or terminal bronchioles)
predominantly adenocarcinomas
- Begins insitu in mucosa , grow in to the lumen :
intraluminal mass ; penetrate through the wall in to
peribronchial tissue
- Dx : Sputum , BAL ,FNAC
- Extend to pleura & pericardium
- Tracheal , bronchial , mediastinal LNs
- Metastasize early (except SCC) , adrenals > 50%

Squamous cell carcinoma

- Most commonly found in men
- Closely related with smoking history
- Characterized by the presence of keratinization and intercellular bridges
- The highest frequency of p53 mutations

- **Adenocarcinoma**

- Glandular differentiation / mucin production
- Pattern of growth : acinar , papillary , bronchialveolar , solid
- Most common in women and non smokers
- Peripherally located, smaller vs SCC
- Grow more slowly , metastasize early vs SCC
- Are less frequently associated with smoking
- K-ras mutations are primarily seen



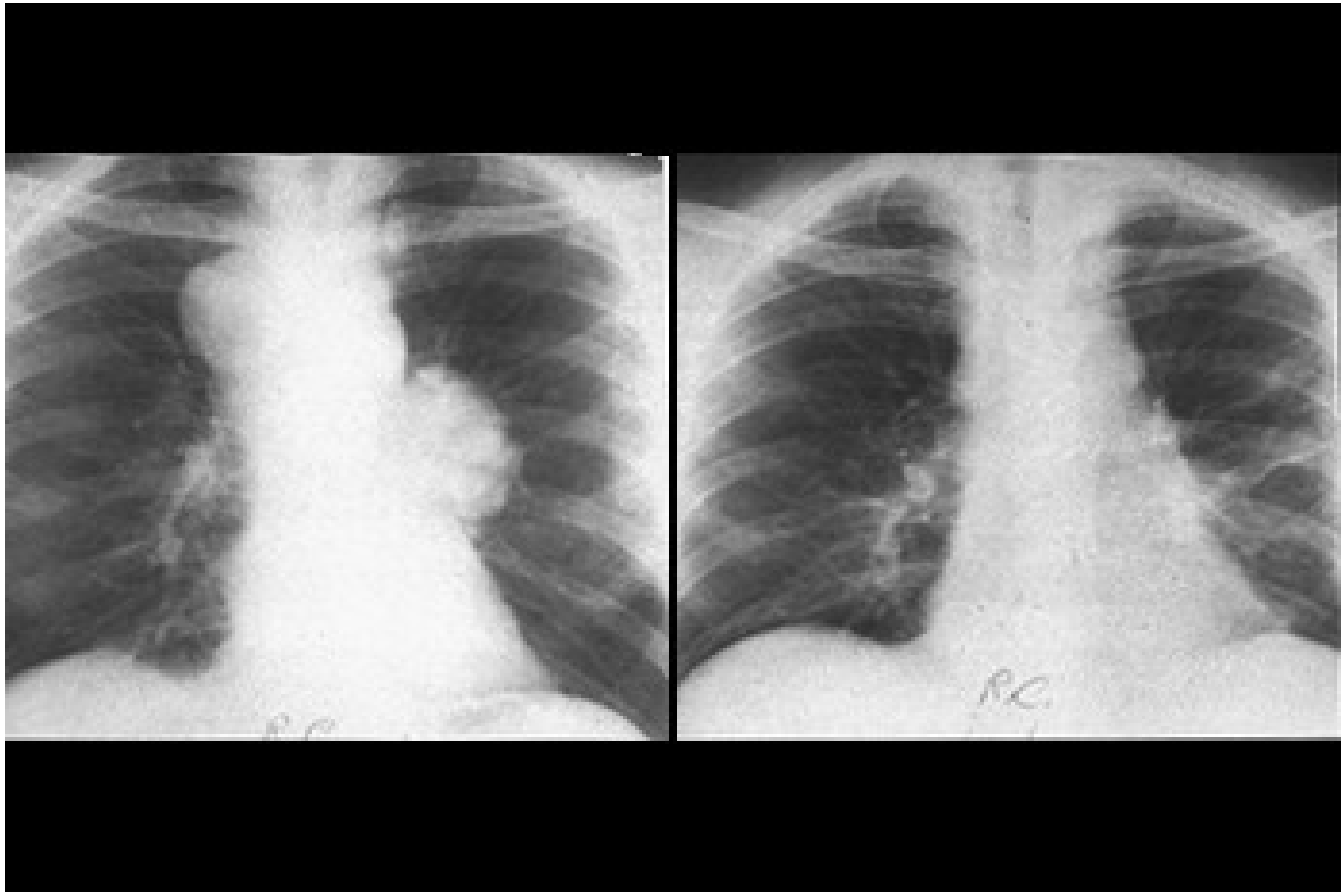
- Lung, squamous cell carcinoma
- Lung, gland-forming adenocarcinoma



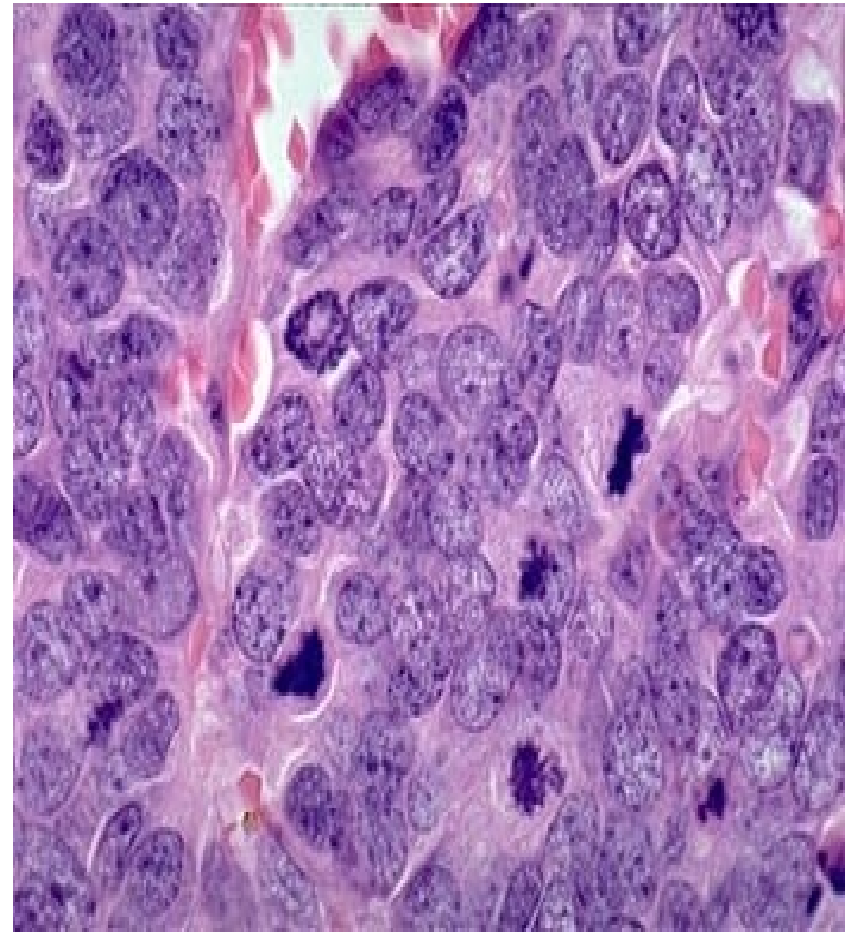
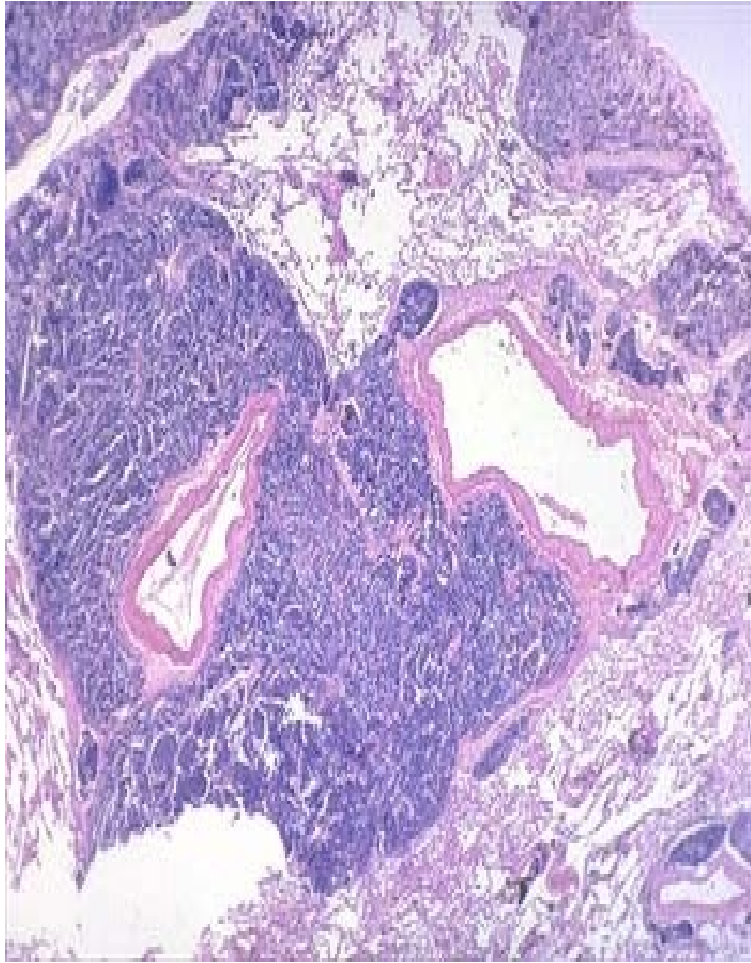
- Fingers, clubbing - Clinical presentation

- **Small cell carcinoma**

- Epithelial cells are small with scant cytoplasm
- Cells are round, oval, and spindle shaped, and nuclear molding is prominent
- Mitotic count is high
- Necrosis is common
- High grade tumor
- Most common pattern associated with ectopic hormone production
- Have a strong relationship to cigarette smoking
- Occur in both major bronchi and periphery of the lung
- The most aggressive of lung tumors



- Chest, right lung mass - Radiographs



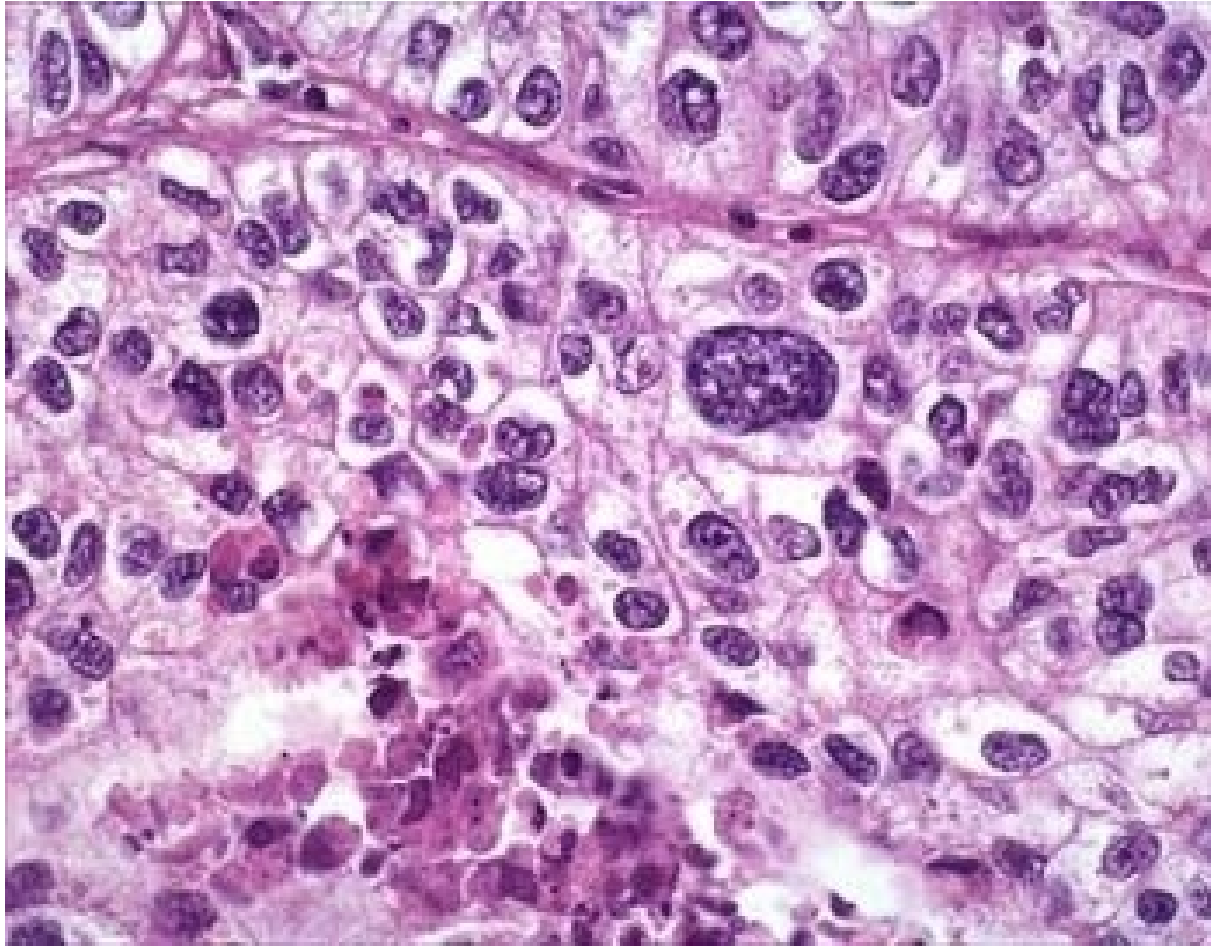
- small cell carcinoma

- **Large cell carcinomas**

- Undifferentiated malignant epithelial tumor
- Cells typically have large nuclei, prominent nucleoli, and a moderate amount of cytoplasm
- Large cell neuroendocrine carcinoma , histologic variant
- 10 % have combined histology

• **Secondary pathology**

- **Partial obstruction** – cause marked focal emphysema
- **Total obstruction** may lead to atelectasis
- **Impaired drainage of the airways** - severe suppurative or ulcerative bronchitis or bronchiectasis
- Pulmonary abscesses
- **Superior venacava syndrome** (compression/invasion)
 - Venous congestion, dusky head, arm edema, circulatory compromise
- **Effusions** – pericarditis , pleuritis



- Lung, large cell carcinoma

- **Clinical course**

- Cough, Weight loss, Chest pain, Dyspnea
- Metastatic tumors
- Paraneoplastic syndromes
 - Are commonly seen with lung carcinomas
- Apical lung cancers (**pancoast tumors**) tend to invade neural structures around the trachea(cervical sympathetic plexus)=
Hornor syndrom + pain along distribution of ulnar nerve
- Outlook is poor
- 5 year survival rate is 15%

Neuroendocrine tumors

- Neuroendocrine cells with in the epithelium
- Benign tumorlets - small , benign , hyperplastic , in association with fibrosis/inflammation
- Carcinoids
- Small & large cell neuroendocrine carcinomas
- They Share Morphologic, ultra structural, molecular genetics, IHC characteristics ; but significant difference incidence, clinical, epidemiologic, histologic, survival.

Carcinoid tumors

- 1 - 5% of lung tumors , most younger than 40
- Equal incidence in both sexes
- Low grade malignant epithelial neoplasms
- Typical and atypical carcinoids - morphology
- Morphology : central / peripheral
- Central : finger like / spherical polypoid masses , project in to the lumen , covered by intact mucosa
- 3 - 4 cm , most confined to stem bronchi.

Metastatic tumors

- most common site of metastatic neoplasm
- BV, lymphatic , direct continuity
- Variabl pattern of growth
- Multiple discrete nodules(cannonball lesions)
throughout all lobes - usual
- Tend to be peripheral

- **Pleural diseases**

- **Pleural effusion :**

- **Inflammatory pleural effusions**

- From pleuritis
 - Common causes are tuberculosis, pneumonia, lung infarcts, lung abscess, bronchiectasis
 - Other causes of pleuritis : SLE , metastasis
 - **Empyema** – usually from bacterial or mycotic seeding

- **Non inflammatory pleural effusions**

- Hydrothorax
 - Hemothorax
 - Chylothorax

- **Pneumothorax**

- Air or gas in the pleural cavities
- Spontaneous, traumatic, therapeutic
- Most commonly associated with emphysema, asthma, and tuberculosis
- Causes compression, collapse, and atelectasis of the lung and may be responsible for marked respiratory distress

- **Pleural tumors**

- **Secondary tumor** is more common than primary tumors
- Lung and breast most common , any organ
- Serous / serosanguineous effusion follows

- **Solitary fibrous tumors**

- Attached to pleural surface
- No relationship to asbestos exposure
- No effusion
- Remain confined to the lung

- **Malignant Mesothelioma**

- Heavy exposure to asbestos
- Associated with extensive pleural effusion
- Affected lung gets ensheathed by tumor tissue
- Epitheloid or mesenchymal type of cells
- Clinical course
 - Chest pain
 - Dyspnea
 - Recurrent pleural effusions



- Lung, mesothelioma - Gross